An Inquiry into Multiple Spatial Dimensions

A THESIS

By

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Introduction

Higher dimensional concepts occur in a variety of cutting edge physical and mathematical contexts. Higher dimensional spaces occur frequently in theoretical physics in the form of configuration spaces for Lagrangians and Hamiltonians. String theory predicts our universe requires eleven additional spatial dimensions for physical equations from symmetry principles and internal consistency. Higher dimensional arrays occur routinely in computer science. General relativity requires the use of tensors in a four dimensional space-time manifold. And there are many more examples.

However it is very difficult to get a physical intuition for these spaces. For instance, given unit radii, the hypervolume of an eight dimensional sphere is less than that of the hypervolume of a seven dimensional sphere, but that of a five dimensional hypersphere is greater than that of a four dimensional hypersphere. This is a surprising result! Recursion is a very broad tool and can be seen as a unifying framework under which many phenomena can be understood in terms of previous phenomena or simpler concepts. The approach in this paper is to understand these spaces and operators using only our intuitive ideas of lower (i.e. two or three) dimensional examples, and using a recursive method to define higher dimensional cases. The higher dimensional cases brought forth in this paper are very simple higher dimensional objects, the objects used at the edge of mathematical research being predicated on compositions of these and other concepts in more detail.

Generating functions

This first chapter deals primarily with the generating functions for simplices and hypercubes, which are some simple examples of discrete higher dimensional objects. Next I compute generating functions for mixed simplex-hypercubes. Last is a computation of the hyperareas and hypervolumes of these figures.

To figure out the general rules for constructing a simplex it is instructive to look at the lower dimensional cases. For example, a triangle is constructed by taking a line segment, finding the middle, moving away from it in the second dimension and making a point. Then, one connects the other points to the new point with a line segment. For a tetrahedron one starts with a triangle, finds its center, and projects from that point orthogonally into the 3^{rd} dimension. As before this new point is connected to the other points by means of line segments. In general the rule is to add a point in a new direction. It must be orthogonal to the other directions, and directly above the center of the previous figure. Once such a point is made, connect all the old points to the new point with line segments. This process is shown for the first few iterations in the following Figure 1. The points are numbered so that the reader can track the changes that occur from one figure to the next.

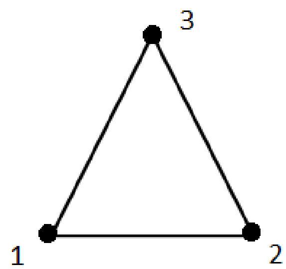
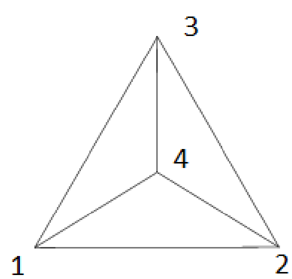
 	This is a point
 	Two points One line segment
 	Three points Three line segments One triangle
 	Four points Six line segments Four triangles One tetrahedron

Figure 1: *0, 1, 2, and 3 simplices (Wolfram)*

The rules for generating hypercubes can also be inferred from the lower dimensional versions. When we want to create a square from a line segment we can simply translate a copy of the line segment across in the second dimension and connect the new points to the ones that they were copied from to make a square. To make a cube from a square the procedure is similar: we translate a copy of the entire square in the 3^{rd} dimension, then connect the points to the original points. The general rule then is to copy the figure into the next orthogonal dimension and connect the points across in that dimension. This is shown in the Figure 2. As before the points are numbered so inferences can be drawn across dimensions.

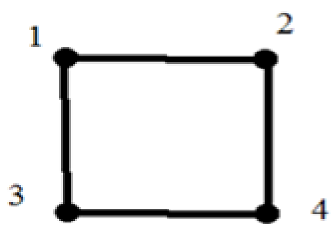
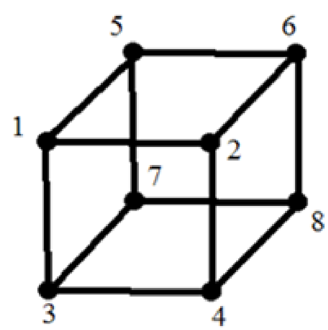
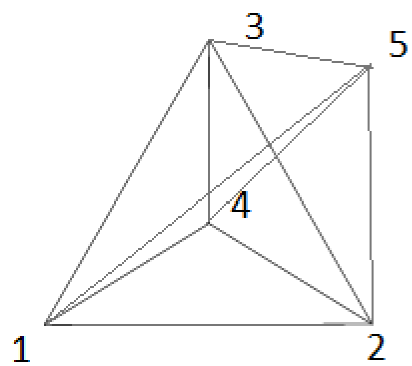
	This is a point
	Two points One line segment
	Four points Four lines One square
	Eight points Twelve lines Six squares One cube

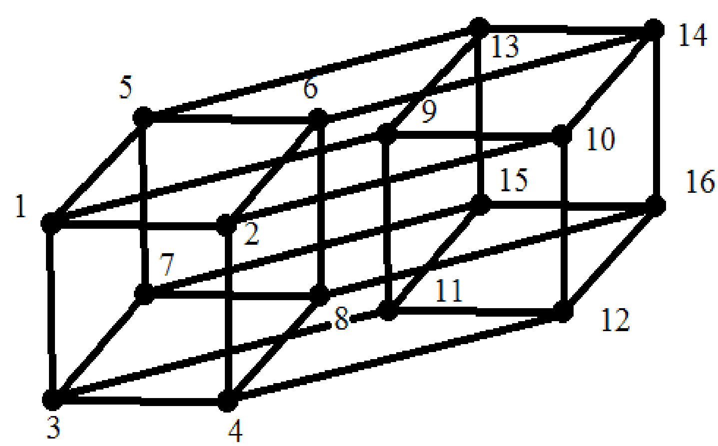
Figure 2: *0, 1, 2, and 3 dimensional hypercubes*

We run out of figures that are easily rendered however. This is shown in Figure 3 and Figure 4 as the first four dimensional simplex and hypercube. This leads to the use of generating functions to give structure to the higher dimensional versions of the square and triangle.



	Five points Ten line segments Ten triangles Five tetrahedrons One 4 simplex

Figure 3: *The 4 simplex (Wolfram)*



	Sixteen points Thirty-two lines Twenty-four squares Eight cubes One 4 dimensional hypercube

Figure 4: *The 4 dimensional hypercube*

Generating functions are a way of using polynomials and their properties to help collect and organize objects. They are a formal power series with one or more variables and are often used in combinatorics. In this chapter generating functions are used to count the numbers of different dimensional objects in higher dimensional objects.

A formal power series is represented as

$$\sum_{n=0}^{\infty}\left(a_n x^n\right)$$

The number multiplier, or coefficient, in the polynomial tells how many of the objects there are, the power of the variable or indeterminate (x or y) tells the dimension of the object being counted. The coefficient here is a_n and refers to the number of the objects there are, the x^n or y^n correspond to an object with n dimensions. In this paper we assign to powers of the variable x figures associated with simplices, and to powers of the variable y figures associated with hypercubes.

For a few examples consider a triangle, a square and a cube. The triangle is a figure that has one 2 dimensional object, the triangle itself, three 1 dimensional objects which are line segments, and three 0 dimensional objects that are points. This in itself could be used to describe it but we can write it in a more efficient manner as (using the variable x for simplicial figures)

$$x^2\ (one\ triangle) +\ 3x\ (3\ lines) +\ 3\ (3\ points)$$

We can remove the parenthetical descriptions to leave the bare algebraic structure:

$$x^2 + 3x + 3$$

For squares, a very similar notation exists. However squares have four points and four lines and so will look like this:

$$y^2\ (one\ square)\ +\ 4y(4\ lines)\ +\ 4\ (4\ points)$$

We can remove the redundant description here as well:

$$y^2 + 4y + 4$$

Cubes are different from squares and triangles in that they are 3 dimensional. This will correspond to having more objects to count. We can elucidate a cube as:

$$y^3(one\ cube) + 6y^2(six\ squares) + 12y(twelve\ lines) + 8(eight\ points)$$

or $y^3 + 6y^2 + 12y + 8$

The n dimensional simplex will be represented as S_n, the variable will be in x, and it will have *n+1* points. A triangle is a 2 simplex

$$S_{n+1}=S_n$$

because the previous figure is part of the new figure,

$$+x*S_n$$

all points make lines when connected to the new point, all lines form triangles when connected to the new point and all n-1 dimensional figures will likewise form n dimensional figures,

$$+1$$

for the new point since it is not covered in the previous steps.

[This is explained graphically in Figures 1 and 3 and logically in the second paragraph.]

Rewritten and without the explanations it looks like

$$S_{n+1}=(x+1)S_n + 1$$

$$S_0= 1$$

So then we can see that S_1 is a line by

$$x + 2 = (x + 1) * 1 + 1$$

and that S_2 is a triangle

$$x^2 + 3x + 3 = (x + 1)^2 + (x + 1) + 1$$

and that S_n can be written as a sum like so:

$$S_n = \sum_{i=0}^{n} (x + 1)^i$$

We can note that we can also write it via the geometric series as

$$\frac{((x+1)^{n+1} - 1}{(x+1) - 1} = \frac{((x+1)^{n+1} - 1}{x}$$

and since we can reduce this with the binomial theorem

$$(x + 1)^{n+1} = \sum_{k=0}^{n+1} \binom{n+1}{k} x^k 1^{n+1-k}$$

we can write it in a final form as

$$\frac{(x+1)^{n+1} - 1}{x} = \sum_{k=1}^{n+1} \binom{n+1}{k} x^{k-1}$$

A similar analysis will now be applied to hypercubes. The hypercube in n dimensions will be represented as H_n and the variable will be in y. [Reference Figures 2 and 4 for visualization, and paragraph 3 for the logic.] Thus we have

$$H_{n+1} = 2 * H_n$$

because of the doubling in a direction orthogonal will keep the original and make a copy,

$$+y * H_n$$

connecting the points results in new lines, all original lines lead to squares and in general all *n-1* dimensional figures will create *n* dimensional figures.

It can be rewritten as

$$H_{n+1} = (y+2)*H_n$$

and since

$$H_0=1$$

it is easily seen that

$$H_n=(y+2)^n$$

which is the same as the sum given by the binomial theorem:

$$\sum_{i=0}^{n}\binom{n}{i}y^i 2^{n-i}$$

These previous geometric forms had a great deal of symmetry due to being created by only one geometric operation. We can however mix these two types of geometric generating actions. Quickly we come to the conclusion that they do not commute. As an example, start with a line segment. It is denoted as before equally well by $x + 2$ or by $y + 2$. We will choose to represent it as $x + 2$. As the next step we take two distinct copies of the line, one to operate on with a simplex operation, and the other with a hypercube operation. We get both

$$x^2 + 3x + 3$$

and

$$xy + 2y + 2x + 4$$

The interesting part is when we do one of the hypercube operations on the triangle, and one of the simplex operations on the square. The first of these proceeds as

$$(x^2 + 3x + 3)(y + 2)$$

which results in

$$x^2y + 3xy + 3y + 2x^2 + 6x + 6$$

Intuitively when we move a triangle in a space orthogonal to it and connect it across we should have a triangular prism. We now can render our new shape into English as

$$x^2y\ (triangular\ prism) +\ 3xy(3\ square\ faces) +\ 3y\ (3\ lines) + 2x^2(2\ triangles)\ \ +\ 6x\ (6\ mor$$

There is a corresponding illustration in Figure 5

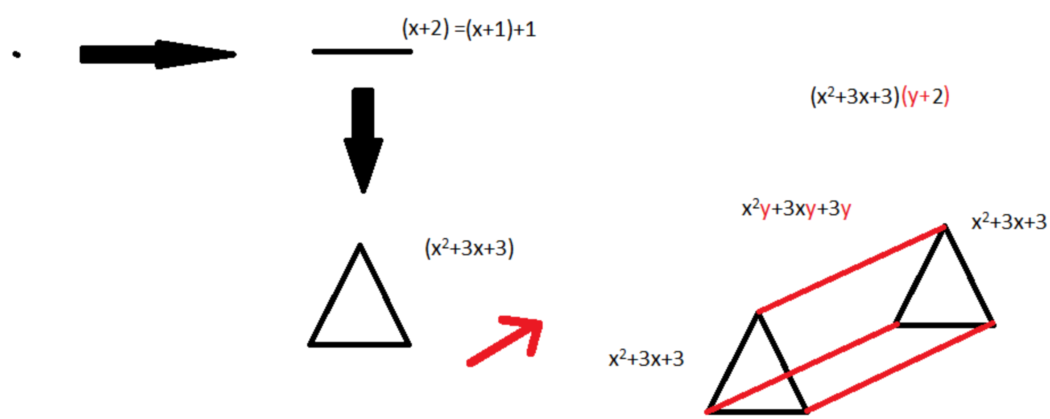


Figure 5: *Triangular prism as an archetype of a mixed figure*

But we still have the square to do a simplex operation on to compare it with:

$$(xy + 2y + 2x + 4)(x + 1) + 1$$

$$= xyx + 2yx + 2x^2 + 4x + xy + 2y + 2x + 5$$

And if we think about it intuitively, if we have a square and then add one more point above it and connect all the other points we get a square pyramid figure

$(xyx\ (square\ pyramid) + 2yx(2\ triangles) + 2x^2\ (2\ triangles) + 4x\ (4\ lines) + xy(\ square) +$

and we can compare differences between these two figures. One has an xy contribution, the other has a yx contribution, and we can see that one results in squares, the other triangles. Neither the operations nor the variables will commute.

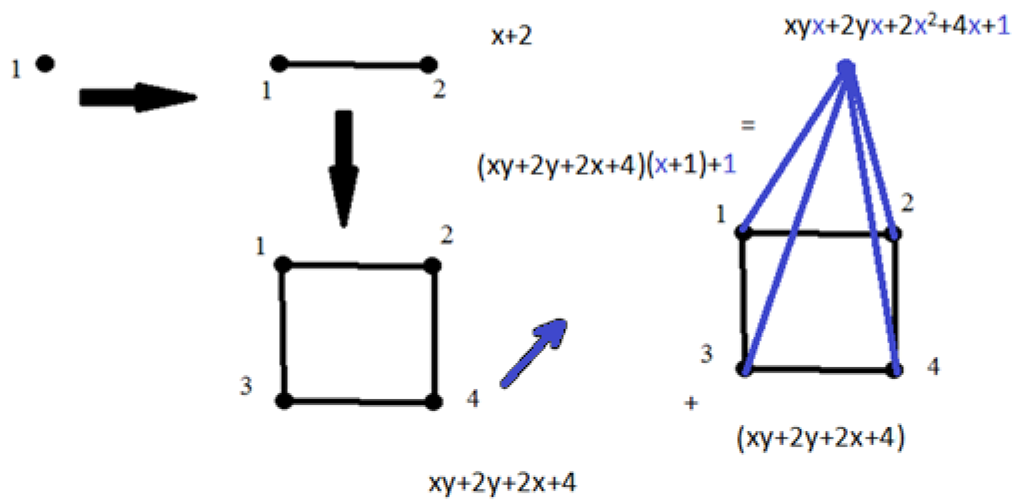


Figure 6: *Square pyramid as an archetype of a mixed figure*

Figures 5 and 6 have arrows with two different colors to better display the changes a figure undergoes when being translated orthogonally or by being connected to an point lying orthogonal to it

But why stop here? After all there are higher dimensional versions of these 3 dimensional mixed figures. We can label each figure by a vector in a higher dimensional discrete noncommutative function space. This helps organize things by telling what operations have occurred to create the figure.

The method I use utilizes a vector in the aforementioned function space whose odd indices indicate steps of the hypercube type, and whose even terms indicate steps of the simplex type. For an example, (1,4,7,2) would correspond to starting with a point, doing one hypercube operation, then 4 simplex operations back to back, then 7 hypercube operations, then 2 simplex operations to create a 14 dimensional figure.

For a specific example if we have a vector of this sort

$$\left(a_1, b_1, a_2, b_2, a_3, b_3\right)$$

it corresponds to

$$(((y + 2)^{a_1}(x + 1)^{b_1} + \sum_{i=0}^{b_1-1}(x + 1)^i)(y + 2)^{a_2}(x + 1)^{b_2} + \sum_{i=0}^{b_2-1}(x + 1)^i)(y + 2)^{a_3}(x + 1)^{b_3} + \sum_{i=0}^{b_3-1}(x + 1)^i$$

which looks daunting. If we take it term by term though it is much simpler. Also a simplification by using different letters to show what is at the core of the calculation is quite helpful. If we substitute A_i for $(y + 2)^{a_i}$ and $(x + 1)^{b_i}$ with B_i and lastly $\sum_{j=0}^{b_i-1}(x + 1)^j$ with C_i it will look much nicer. Explicitly,

$$(((y + 2)^{a_1}(x + 1)^{b_1} + \sum_{i=0}^{b_1-1}(x + 1)^i)(y + 2)^{a_2}(x + 1)^{b_2}$$

$$+ \sum_{i=0}^{b_2-1}(x+1)^i)(y+2)^{a_3}(x+1)^{b_3} + \sum_{i=0}^{b_3-1}(x+1)^i$$

$$= \left(\left(A_1 * B_1\right) + C_1\right)\left(A_2 * B_2\right) + C_2\right)\left(A_3 * B_3\right) + C_3\right)$$

$$= (A_1 * B_1)(A_2 * B_2)\left(A_3 * B_3\right) + C_1(A_2 * B_2)\left(A_3 * B_3\right) + C_2(A_3 * B_3) + C_3$$

This can be written as

$$\prod_{k=1}^{3}(A_k * B_k)) + \sum_{j=1}^{3}((C_j) * \prod_{k=j+1}^{3}(A_k * B_k))$$

or, if we allow there to exist an initial C_0 that equals 1 we can write it as a sum of products, useful for disentangling nested sequences,

$$\sum_{j=0}^{3}((C_j) * \prod_{k=j+1}^{3}(A_k * B_k))$$

We come to realize that our large sum that corresponds to our vector can be rewritten in a simpler and more elegant form, namely that the vector

$$\left(a_1, b_1, a_2, b_2 \ldots a_n, b_n\right)$$

will correspond to a figure described by

$$\sum_{j=0}^{n}((C_j) * \prod_{k=j+1}^{n}(A_k * B_k))$$

We can prove this result inductively. Using a base case of n=1,

$$\sum_{j=0}^{1}((C_j) * \prod_{k=j+1}^{1}(A_k * B_k))$$

$$= \sum_{j=0}^{1}((C_j) * (A_1 * B_1))$$

$$= C_1 + (A_1 * B_1)$$

$$= \sum_{j=0}^{b_1-1} ((x+1)^j + (y+2)^{a_1}(x+1)^{b_1}$$

as intuitively expected. We next assume it holds for *n* and prove it will hold for *n+1* from that:

$$\sum_{j=0}^{n} ((C_j) * \prod_{k=j+1}^{n} (A_k * B_k))$$

Then if we apply hypercubing a_{n+1} many times followed by b_{n+1} many simplexing steps:

$$\sum_{j=0}^{n} ((C_j) * \prod_{k=j+1}^{n} (A_k * B_k))A_{n+1} B_{n+1} + C_{n+1}$$

$$= \sum_{j=0}^{n+1} ((C_j) * \prod_{k=j+1}^{n+1} (A_k * B_k))$$

as expected.

If we substitute back what our capital letters stood in for we get (remembering that b_0=1)

$$\sum_{j=0}^{n} (\sum_{q=0}^{b_j-1} (x+1)^q) * \prod_{k=j+1}^{n} ((y+2)^{a_k}(x+1)^{b_k}))$$

which from our previous analysis can be further decomposed into

$$\sum_{j=0}^{n} (\sum_{k=1}^{b_j} \binom{b_j}{k} x^{k-1}) * \prod_{k=j+1}^{n} ((\sum_{i=0}^{a_k} \binom{a_k}{i} y^i 2^{a_k - i})(\sum_{i=0}^{b_k} \binom{b_k}{i} x^i))$$

These figures have measures associated to them that generalize the concept of volume and area. In general for a figure that has *n* dimensions it will have a volume measure (called hypervolume) that is *n* dimensional, and spans the entirety of the space much like our intuitive ideas of volume. It will also have an area measure (called hyperarea) that is *n-1* dimensional. Like regular area it corresponds to notions of measuring the size of the boundary covering the volume. For the following discussion all figures are assumed to be regular, that is every edge is of the same length.

For all the above mentioned figures the measures can be derived with elementary calculus. The hypervolume in an *n* dimensional hypercube can be calculated by integrating from 0 to the side length (s) for each dimension.

$$Hypervolume\ of\ n\ hypercube = \int_0^s \int_0^s \dots \int_0^s \prod_{i=1}^{n} dx_i$$

$$= \left(\int_0^s \prod_{i=1}^{n} dx_i\right)^n = s^n$$

This, due to the orthogonality, ends up being simply multiplying the side length of any side by itself *n* times.

Hyperarea of the *n* dimensional hypercube is calculated by integrating the hyperarea element of the previous *n-1* dimensional figure from 0 to *s*, and adding twice the previous *n-1* dimensional hypercubes hypervolume. For lower dimensional examples of why this is so consider finding the perimeter of a square, and the area of a cube. To find the perimeter of a square we integrate the two points bounding a line from 0 to *s*, and add twice the length of the line segment. [For visualization consult Figure 1.]

$$Perimeter = \int_0^s 2dx_2 + 2s = 4s$$

For a cube we integrate the perimeter of the square from 0 to *s*, and add twice the area of the original square

$$Area = \int_0^s 4sdx_3 + 2(s^2) = 6s^2$$

We can write it in a recursive form as

$$Hyperarea\ of\ n\ dimensional\ hypercube = HA_n = \int_0^s HA_{n-1}dx_n + 2s^{n-1}$$

$$= (\int_0^s (\int_0^s HA_{n-2} dx_{n-1}) + 2s^{n-2} dx_n) + 2s^{n-1}$$

$$= (\int_0^s (\int_0^s HA_{n-2} dx_{n-1}) dx_n) + 4s^{n-1}$$

$$= (\int_0^s (\int_0^s \int_0^s HA_{n-3} dx_{n-2}) dx_{n-1}) dx_n) + 6s^{n-1}$$

$$= 2n * s^{n-1}$$

Simplices are not as simple to analyze since their sides are not all orthogonal to one another. When we integrate them the variable actually shows up in the integrand, not just constants.

As an example, consider computing the area of a regular triangle. We will wish to integrate the line segment with a side length of *a* from 0 to the height of the triangle, denoted *b*. At a distance of *b* away from the apex the length across is *a*, and at a distance of 0 from the apex the length across is zero. Since it varies linearly we now have our scaling factor:

$$\int_0^b a(\frac{h}{b})^1 dh = \frac{ab}{2}$$

This gives the correct formula for the area of a triangle. Let's try it for a tetrahedron. Given a triangular base of area $\frac{ab}{2}$ and an orthogonal height of *c* from the triangle, we integrate the base over the appropriate scaling factor from 0 to *c*

$$\int_0^c \frac{ab}{2}\left(\frac{h}{c}\right)^2 dh = \frac{abc}{6}$$

The general principle involved can be cast in a recursive form as before and solved explicitly in terms of the heights h_i. The volume of an *n* simplex is computed as an integral over the volume of an *n-1* simplex times an appropriate scaling factor

$$HV_n = \int_0^{h_n} HV_{n-1}\left(\frac{h}{h_n}\right)^{n-1} dh = \frac{h_n}{n}HV_{n-1} = \frac{h_n}{n}\frac{h_{n-1}}{n-1}HV_{n-2}$$

$$= \frac{\prod_{i=1}^{n} h_i}{n!}$$

From our previous discussion we know all regular *n*-simplices have *n* + 1 congruent *n-1* simplices as faces. We can use each face as a base, and connect them with lines to the center of the figure to make *n+1* smaller congruent *n* simplices. We obtain an expression for V_n in terms of the apothem a_n (the radius of the hypersphere inscribed in the *n*-simplex),

$$HV_n = \frac{h_n}{n}HV_{n-1}$$

$$HV_n = (n + 1)\frac{a_n}{n}HV_{n-1}$$

$$h_n = (n + 1)a_n$$

Using this we may now find an explicit formula for h_n. In general we can set up a triangle in the middle of the *n* simplex and with the Pythagorean Theorem as

$$h_{n-1}^2 = h_n^2 + a_{n-1}^2$$

We can derive this with analogy to the tetrahedron. For the triangle, $h_2 = \frac{\sqrt{3}}{2}s$. The height h_3 of the tetrahedron is found by first knowing the height of the triangle, for the line segment (of length h_3) from a vertex of the tetrahedron to the center of the opposite face makes a right triangle with the slant height segment of the pyramid (of length h_2) and the apothem segment of the base (of length a_2). This is illustrated as Figure 7. The blue figure represents an *n-1* dimensional simplex, detail has been omitted to show the essential parts.

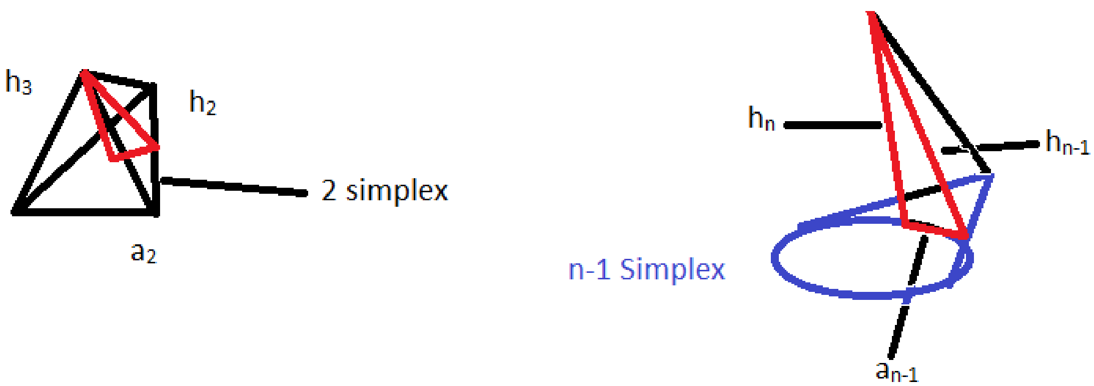


Figure 7: *Cross-dimensional analogy between triangle and n-1 simplex for finding apothem and height*

We can use the fact that $\frac{h_n}{(n+1)} = a_n$ to derive a recursion relation

$$h_{n-1}^{\ 2} = h_n^{\ 2} + {\frac{h_{n-1}}{n}}^2$$

$$h_{n-1}^{\ 2} - {\frac{h_{n-1}}{n}}^2 = h_n^{\ 2}$$

$$h_n^{\,2} = h_{n-1}^{\,2}(1 - 1/n^2)$$

$$h_n^{\,2} = h_{n-1}^{\,2}\left(\frac{n^2-1}{n^2}\right)$$

$$h_n = \frac{\sqrt{n^2-1}}{n} h_{n-1}$$

Since $h_2 = \frac{\sqrt{3}}{2}s$,

$$h_n = \sqrt{\frac{n+1}{2n}}\, s$$

$$V_n = \frac{h_n V_{n-1}}{n} = \frac{1}{n}\sqrt{\frac{n+1}{2n}}\, sV_{n-1}$$

We can get an explicit measure with the recursion relation; $V_1=s$, since it is the 1 dimensional volume of a line. The $1/n$ contribution leads to a factorial, the square roots of two lead to n / 2 many powers of 2 and each square root of n + 1 will cancel out the square roots of n except for the last one. Thus

$$V_n = \frac{\sqrt{\frac{n+1}{2^n}}}{n!} s^n$$

is the hypervolume of an n dimensional simplex.

Hyperspheres' Hypervolumes

This chapter concerns itself with hyperspheres and their hypervolumes and hyperareas. Hyperspheres are the natural higher dimensional generalization of circles and spheres. They occur when one tries to find the set of points equidistant from a central point in a higher dimensional Euclidean space. When trying to find the higher dimensional volume one must resort to doing multiple integrals, *n* integrals for *n* many dimensions. Symbolically this can be represented as

$$\int_{R^n} dV = V_n$$

where the single integral of R^n stands for the *n* dimensional space it is integrated over in *n* dimensions, and dV stands for the volume element. When we evaluate this explicitly in Euclidean coordinates it becomes

$$\int_{-r}^{r} \int_{-\sqrt{r^2-x^2}}^{\sqrt{r^2-x^2}} \dots \int_{-\sqrt{r^2-\Sigma x_i^2}}^{\sqrt{r^2-\Sigma x_i^2}} \prod_{i=1}^{n} dx_i$$

which is slightly unpleasant to evaluate. Fortunately if we switch to a coordinate system that uses the angular symmetry of the system it becomes much easier. The system of coordinates we will use to evaluate it are called hyperspherical coordinates. Instead of using the directions up down, left right, forward and back to define the coordinates a different orthogonal system is used. It uses the radial coordinate, which tells how far away from the center the point is, and a set of angles each orthogonal to one another.

But before we address the general coordinates it is helpful to start with the 2 dimensional version, finding the area of a circle using the polar coordinate system. One way of defining it is shown in Figure 8.

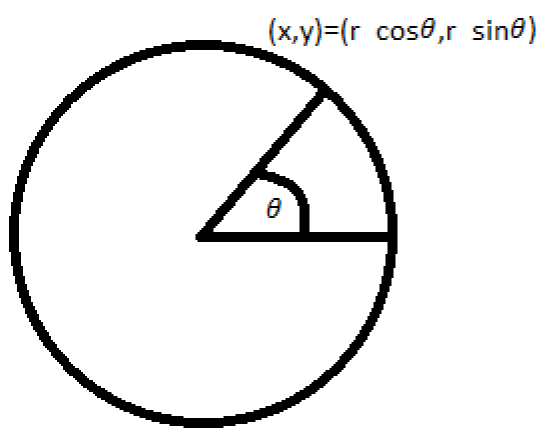


Figure 8: *A circle in polar coordinates parameterized by r and* θ

$$x = r \cos cos \, \theta$$

$$y = r \sin sin\,\theta$$

This makes

$$x^2 + y^2 = r^2$$

which defines an equidistant surface for constant *r*, the common circle.

In order to use the new coordinate system we have to account for the fact that the volume element is different. The way we do so is by taking the determinant of a special matrix called the Jacobian. The Jacobian is defined for switching from Cartesian to polar coordinates as

$$\frac{\partial(x,y)}{\partial(r,\theta)} = \left[\frac{\partial x}{\partial r} \;\; \frac{\partial x}{\partial \theta} \;\; \frac{\partial y}{\partial r} \;\; \frac{\partial y}{\partial \theta} \right]$$

Which is explicitly

$$[\cos cos\,\theta \quad - r \sin sin\,\theta \quad \sin sin\,\theta \;\; r \cos cos\,\theta \;]$$

Note that we can factor this into two different matrices. This trick comes up later in the analysis and helps limit the number of computations

$$\cos cos\,\theta \quad - r \sin sin\,\theta \quad \sin sin\,\theta \;\; r \cos cos\,\theta \quad = \left[\frac{x}{r} \;\; - \;\theta \;\; \frac{y}{r} \;\theta \right] = (x \; 0 \; 0 \; y)\left[\frac{1}{r} \;\; - \tan tan\,\theta \;\; \frac{1}{r}\right.$$

When we take the determinant of the Jacobian it is equivalent to taking the determinant of the second matrix times each position, in this case x*y. This is because each term in the determinant of the Jacobian has a factor of x*y at the end. The determinant of the polar Jacobian is equal to *r*.

The next step is to change the boundaries of the integration and the volume element to be in terms of the new coordinates. This means turning

$$\int_{-r}^{r} \int_{-\sqrt{r^2-x^2}}^{\sqrt{r^2-x^2}} dx * dy$$

into

$$\int_{0}^{r} \int_{0}^{2\pi} \det det\,(J) \; * dr * d\theta$$

$$\int_{0}^{r} \int_{0}^{2\pi} r * dr * d\theta$$

This is much simpler to compute. It is also separable into the product of two independent integrals such that

$$\int_{0}^{r} r * dr \int_{0}^{2\pi} d\theta = \pi r^2$$

Consider the more general case of an *n* dimensional hypersphere. As before we first set up a coordinate system so that the integral will be simpler. Let us label the Euclidean coordinates by x_i and the hyperspherical ones by *r* and by θ_i .This is given by

$$x_1 = r\cos cos\,\theta_1$$

$$x_2 = r\sin sin\,\theta_1 \; \cos cos\,\theta_2$$

$$x_3 = r\sin sin\,\theta_1 \; \sin sin\,\theta_2 \; \cos cos\,\theta_3$$

$$\vdots$$

For all $j<n$

$$x_j = r\cos cos\,\theta_j$$

$$\vdots$$

$$x_{n-1} = r \cos cos\, \theta_{n-1}$$

$$x_n = r \prod_{i=1}^{n-1} \sin sin\, \theta_i$$

These coordinates are chosen so that

$$\sum_{i=1}^{n} (x_i)^2 = r^2$$

Next is computing the determinant of the Jacobian. The Jacobian is

$$\frac{\partial(x_1, x_2, x_3 \dots x_n)}{\partial(r, \theta_1, \theta_2 \dots \theta_{n-1})} = \Big(\frac{\partial x_1}{\partial r}\ \frac{\partial x_1}{\partial \theta_1}\ \frac{\partial x_1}{\partial \theta_2}\ \frac{\partial x_2}{\partial r}\ \frac{\partial x_2}{\partial \theta_1}\ \frac{\partial x_2}{\partial \theta_2}\ \frac{\partial x_3}{\partial r}\ \frac{\partial x_3}{\partial \theta_1}\ \frac{\partial x_3}{\partial \theta_2}\ \cdots\ \frac{\partial x_1}{\partial \theta_{n-2}}\ \frac{\partial x_1}{\partial \theta_{n-1}}\ \frac{\partial x_2}{\partial \theta_{n-2}}\ \frac{\partial x_2}{\partial \theta_{n-1}}\ \frac{\partial x_3}{\partial \theta_{n-2}}\ \frac{\partial x_3}{\partial \theta_{n-1}}$$

$$= \Big(\frac{x_1}{r}\ - x_1 \tan tan\, \theta_1\ 0\ \frac{x_2}{r}\ \cot x_2 cot\, \theta_1\ - \tan x_2 tan\, \theta_2\ \frac{x_3}{r}\ x_3 \cot cot\, \theta_1\ x_3 \cot cot\, \theta_2\ \cdots\ \vdots\ \ddots\ - \tan$$

which can be evaluated explicitly as the product of two matrices:

$$\big(x_1\ 0\ 0\ 0\ x_2\ 0\ 0\ 0\ x_3\ \cdots\ 0\ \vdots\ \ddots\ \vdots\ 0\ \cdots\ x_{n-1}\ 0\ 0\ x_n \big)\Big(\frac{1}{r}\ - \tan tan\, \theta_1\ 0\ \frac{1}{r}\ \cot cot\, \theta_1\ - \tan tan$$

Some explanation may be of help here. There are 4 separate cases to consider in the derivatives.

Case one is the derivative is with respect to the radial coordinate $\frac{\partial x_i}{\partial r}$. What occurs since each Euclidean coordinate is defined with only one power of *r*, is that it is effectively multiplied by 1/*r*.

Case two is the derivative is of an angular coordinate whose index is higher than that of the Euclidean index $\frac{\partial x_i}{\partial \theta_{i+1}}$. Since the Euclidean coordinates are defined (with the exception of *n*) in a way that the highest angular coordinates index is equal to its index, there is no contribution and it is hence multiplied by zero.

Case three is the derivative is of an angular coordinate whose index is equal to the Euclidean coordinates index $\frac{\partial x_i}{\partial \theta_i}$. Since the Euclidean coordinates are defined in such a way that every angular coordinate that matches its Euclidean one is in a factor of $\cos cos\, \theta_i$, the derivative of cosine is negative sine. If we multiply $\cos cos\, \theta_i$ by θ_i then we get θ_i, the contribution we wanted. For clarity this contribution has been highlighted in gray.

Case 4 is the derivative of an angular coordinate whose index is less than the Euclidean coordinate's index $\frac{\partial x_i}{\partial \theta_{i-1}}$. The Euclidean coordinates are defined in such a way that every angular coordinate that is less than its Euclidean one is in a factor of $\sin sin\, \theta_i$, the derivative of sine is cosine. If we multiply $\sin sin\, \theta_i$ by $\cot cot\, \theta_i$ then we get $\cos cos\, \theta_i$, the contribution we wanted.

It can be shown that the determinant of a product of a diagonal matrix and a normal matrix is equal to the product of the determinants of the matrices with an inductive proof:

Let D be a diagonal matrix with entries d_{ii}, and let A be an ordinary matrix with entries a_{ij}. Mathematically we represent the statement as

$$Det(DA) = Det(D) * Det(A)$$

Base case: Both D and A are 1 by 1 matrices (D^1 and A^1). In this case

$$Det\left(D^1A^1\right) = d_{11} * a_{11}$$
$$= Det(D^1) * Det(A^1)$$

which shows it is valid as a one dimensional case.

Induction step: Assume the theorem holds for *n* x *n* matrices. Then prove it will hold for the *n+1* x *n+1* case.

$$Det\left(D^nA^n\right) = Det(D^n) * Det(A^n)$$
$$= Det\left(D^{n+1}A^{n+1}\right)$$

This can be expanded via Cramer's rule to be

$$= \sum_{j=1}^{n+1} d_{n+1,n+1} * a_{n+1,j} Det\left(D^nA^{nj}\right)$$

where A^{nj} is the matrix formed by taking the *n+1*th row and *j*th column out of the matrix A^{n+1}. From our assumption step this can be factored into

$$= \sum_{j=1}^{n+1} d_{n+1,n+1} * a_{n+1,j} Det(D^n) * Det(A^{nj})$$
$$= \sum_{j=1}^{n+1} d_{n+1,n+1} Det(D^n) * a_{n+1,j} * Det(A^{nj})$$
$$= d_{n+1,n+1} Det(D^n) \sum_{j=1}^{n+1} * a_{n+1,j} * Det(A^{nj})$$
$$= Det(D^{n+1}) Det(A^{n+1})$$

The polar coordinate case has already been discussed. For clarity I now introduce the case for n=3 the spherical coordinates.

$$\left(x_1\ 0\ 0\ 0\ x_2\ 0\ 0\ 0\ x_3\right)\left(\frac{1}{r}\ -\tan tan\ \theta_1\ \ 0\ \frac{1}{r}\ \cot cot\ \theta_1\ \ -\tan tan\ \theta_2\ \frac{1}{r}\ \cot cot\ \theta_1\ \ \cot cot\ \theta_2\right)$$

When we take the determinant of the diagonal matrix first we get

$$x_1x_2x_3 * \, Det\left(\frac{1}{r} \quad -\tan tan\,\theta_1 \quad 0 \ \frac{1}{r} \quad \cot cot\,\theta_1 \quad -\tan tan\,\theta_2 \ \frac{1}{r} \quad \cot cot\,\theta_1 \quad \cot cot\,\theta_2 \right)$$

and the second determinant can be evaluated with Cramer’s rule. If we expand around the right hand side we get

$$\cot cot\,\theta_2 \ Det\left(\frac{1}{r} \quad -\tan tan\,\theta_1 \ \frac{1}{r} \quad \cot cot\,\theta_1 \right) + \tan tan\,\theta_2 \, Det\left(\frac{1}{r} \quad -\tan tan\,\theta_1 \ \frac{1}{r} \quad \cot cot\,\theta_1 \right)$$

which is equal to

$$\frac{1}{r} * \ (\cot cot\,\theta_2 \ + \ \tan tan\,\theta_2)\,(\cot cot\,\theta_1 \ + \ \tan tan\,\theta_1)$$

We now multiply these two terms together as

$$x_1x_2x_3 * \frac{1}{r} * \ (\cot cot\,\theta_2 \ + \ \tan tan\,\theta_2)\,(\cot cot\,\theta_1 \ + \ \tan tan\,\theta_1)$$

We can simplify them by substituting the definitions of the x_i's

$$x_1 = r\cos cos\,\theta_1$$

$$x_2 = r\sin sin\,\theta_1 \ \cos cos\,\theta_2$$

$$x_3 = r\sin sin\,\theta_1 \ \sin sin\,\theta_2$$

and by rewriting

$$(\cot cot\,\theta_i \ + \ \tan tan\,\theta_i) \ = \frac{1}{\sin sin\,\theta_i * \cos cos\,\theta_i}$$

so that we come to

$$r^2 * \ \sin sin\,\theta_1$$

In our general case where *n* is arbitrary, the determinants being taken are again

$$\left(x_1\ 0\ 0\ 0\ x_2\ 0\ 0\ 0\ x_3\ \ \cdots\ \ 0\ \vdots\ \ddots\ \vdots\ 0\ \ \cdots\ \ x_{n-1}\ 0\ 0\ x_n\right)\left(\frac{1}{r}\ -\tan tan\,\theta_1\ \ 0\ \frac{1}{r}\ \cot cot\,\theta_1\ \ -\tan tan\right.$$

The first one is simple, as it corresponds to the product of all the x_i's. The second case can be evaluated in an analogous manner to that used in the spherical case. One expands upon the rightmost side recursively with Cramer's rule, and ends up with an expression

$$\frac{1}{r} * \prod_{i=1}^{n-1} (\cot cot\,\theta_i + \tan tan\,\theta_i)$$

$$= \frac{1}{r} * \prod_{i=1}^{n-1} \left(\frac{1}{\sin sin\,\theta_i * \cos cos\,\theta_i}\right)$$

When we multiply the two together and then substitute the definitions of the x_i's we get

$$= \frac{1}{r} * \prod_{i=1}^{n-1} \left(\frac{1}{\sin sin\,\theta_i * \cos cos\,\theta_i}\right) * \prod_{j=1}^{n} x_j$$

$$= \frac{1}{r} * \prod_{i=1}^{n-1} \left(\frac{1}{\sin sin\,\theta_i * \cos cos\,\theta_i}\right) * \prod_{j=1}^{n-1} x_j * x_n$$

$$= \frac{1}{r} * \prod_{i=1}^{n-1} \left(\frac{1}{\sin sin\,\theta_i * \cos cos\,\theta_i}\right) * \prod_{j=1}^{n} \left(r \cos cos\,\theta_j\right) * \ r \prod_{i=1}^{n-1} \sin sin\,\theta_i$$

$$= \frac{1}{r} * \prod_{i=1}^{n-1} \left(\frac{1}{\sin sin\,\theta_i * \cos cos\,\theta_i}\right) * r^n \prod_{j=1}^{n-1} \left(\cos cos\,\theta_j\right)$$

All the cosines will cancel out, the power of *r* is reduced by one, and all the sines are reduced by a power of one as well. This leaves us with

$$= r^{n-1} \prod_{i=1}^{n-1} \left(\sin sin\,\theta_i^{\,n-1-i}\right)$$

as our volume element. All that is left is integration in the new coordinates:

$$\int_0^r \int_0^{2\pi} \int_0^{\pi} \dots \int_0^{\pi} \det det\,(J)\ *\ dr\ *\ \prod_{i=1}^{n-1} d\theta_i$$

$$= \int_0^r \int_0^{2\pi} \int_0^{\pi} \dots \int_0^{\pi} r^{n-1} \prod_{i=1}^{n-1} \left(\sin sin\,\theta_i^{\ n-1-i} \right) *\ dr\ *\ \prod_{i=1}^{n-1} d\theta_i$$

which as before is separable into n separate one dimensional integrals

$$= \int_0^r r^{n-1} dr \int_0^{2\pi} d\theta_{n-1} \prod_{i=1}^{n-2} \int_0^{\pi} \sin sin\,\theta_i^{\ n-1-i} d\theta_i$$

We can do these sine integrals by hand

$$\int_0^{\pi} \sin sin\,\theta^{\ 0} d\theta\ =\ \pi$$

$$\int_0^{\pi} \sin sin\,\theta^{\ 1} d\theta\ =\ 2$$

but since we wish to keep *n* as an arbitrary integer we will need a more powerful method.

We can start by noting that

$$\int_0^{\pi} \sin sin\,\theta^{\ n} d\theta\ = \int_0^{\pi} \sin sin\,\theta^{\ n-2} d\theta\ - \int_0^{\pi} \sin \cos cos\,\theta^{\ 2} sin\,\theta^{\ n-2}\ d\theta$$

and use integration by parts. Here let *u* and *v* be defined such that

$$u\ = \sin sin\,\theta^{\ n-1},\ v\ =\ \ -\ \cos cos\,\theta$$

Then

$$du\ = (n\ -\ 1) * \sin sin\,\theta^{\ n-2} \cos cos\,\theta\ d\theta\ ,\ v\ = \sin sin\,\theta\ \ d\theta$$

and

$$\int_0^{\pi} \sin sin\,\theta^{\ n} d\theta\ = \int_0^{\pi} u\ *\ dv\ = uv]_0^{\pi} - \int_0^{\pi} v\, du$$

$$= -\ \cos cos\,\theta\ \ \sin sin\,\theta^{\,n-1}\Big]_0^{\pi} - \int_0^{\pi} -\ \cos cos\,\theta\,(n-1) * \sin sin\,\theta^{\,n-2} \cos cos\,\theta\; d\theta$$

$$= 0 + (n-1)\int_0^{\pi} \sin sin\,\theta^{\,n-2} \cos cos\,\theta^{\,2} d\theta$$

$$= (n-1)\int_0^{\pi} \sin sin\,\theta^{\,n-2} \theta^{\,2}) d\theta$$

$$= (n-1)\int_0^{\pi} \sin sin\,\theta^{\,n-2} d\theta - (n-1)\int_0^{\pi} \sin sin\,\theta^{\,n} d\theta$$

And it appears that we’ve come to a vicious circle of self-reference. Gathering our wits we realize if we write it out explicitly

$$\int_0^{\pi} \sin sin\,\theta^{\,n} d\theta = (n-1)\int_0^{\pi} \sin sin\,\theta^{\,n-2} d\theta - (n-1)\int_0^{\pi} \sin sin\,\theta^{\,n} d\theta$$

We can collect the terms we want on the left side

$$\int_0^{\pi} \sin sin\,\theta^{\,n} d\theta + (n-1)\int_0^{\pi} \sin sin\,\theta^{\,n} d\theta = (n-1)\int_0^{\pi} \sin sin\,\theta^{\,n-2} d\theta$$

$$n\int_0^{\pi} \sin sin\,\theta^{\,n} d\theta = (n-1)\int_0^{\pi} \sin sin\,\theta^{\,n-2} d\theta$$

$$\int_0^{\pi} \sin sin\,\theta^{\,n} d\theta = \frac{n-1}{n}\int_0^{\pi} \sin sin\,\theta^{\,n-2} d\theta$$

From this we can construct solutions for arbitrary powers of sine given the two integrals we computed above:

$$\int_0^{\pi} \sin sin\,\theta^{\,2} d\theta = \frac{1}{2}\pi$$

$$\int_0^{\pi} \sin sin\,\theta^{\,3} d\theta = \frac{4}{3}$$

And it looks like there are actually 2 solutions depending on whether the integral is of an odd or even sine power. Given the starting conditions and the formula to solve for higher powers in terms of lower ones we can get a more general definition as

$$\{\int_0^{\pi} \sin sin\, \theta^{\,2m+1} d\theta \; = \; 2\prod_{i=1}^{m} \frac{2i}{2i+1} \;\; \int_0^{\pi} \sin sin\, \theta^{\,2m} d\theta \; = \; \pi \prod_{i=1}^{m} \frac{(2i-1)}{2i}$$

But since we have, in general, to compute products of these because of the contribution from

$$\prod_{i=1}^{n-2} \int_0^{\pi} \sin sin\, \theta^{\,i} d\theta$$

we have to consider two cases. Let us do so first for *n=2m*, an even integer. Then

$$= \int_0^{\pi} \sin sin\, \theta^{\,1} d\theta \int_0^{\pi} \sin sin\, \theta^{\,2} d\theta \; ... \int_0^{\pi} \sin sin\, \theta^{\,2m-2} d\theta$$

$$= (2) * \left(\frac{\pi}{2}\right) * \left(2 * \frac{2}{3}\right) * \left(\frac{\pi}{2} * \frac{3}{4}\right) * \left(2 * \frac{2}{3} * \frac{4}{5}\right) * \left(\frac{\pi}{2} * \frac{3}{4} * \frac{5}{6}\right) ...$$

$$= \left(2 \prod_{i=1}^{m-1} \frac{2i}{2i+1}\right) * \;\; \pi \prod_{i=1}^{m} \frac{(2i-1)}{2i}$$

Since we can take each of these integrals in pairs of two as

$$\frac{\pi}{1}, \frac{\pi}{2}, \frac{\pi}{3}, \cdots \frac{\pi}{m}$$

we get π /*i* contribution from each i^{th} pair, that is for all *m*

$$\left(2 \prod_{i=1}^{m-1} \frac{2i}{2i+1}\right) * \;\; \pi \prod_{i=1}^{m} \frac{(2i-1)}{2i} = \frac{\pi}{m}$$

$$\prod_{j=1}^{m-1} \left(2 \prod_{i=1}^{m-1} \frac{2i}{2i+1} * \;\; \pi \prod_{i=1}^{m} \frac{(2i-1)}{2i}\right) = \prod_{j=1}^{m-1} \frac{\pi}{j}$$

$$= \frac{\pi^{m-1}}{(m-1)!}$$

Substituting into our general formula for the volume

$$\int_{R^n} dV = V_n$$

$$= \int_0^r r^{2m-1} dr \int_0^{2\pi} d\theta_{2m-1} \prod_{i=1}^{2m-2} \int_0^{\pi} \sin sin\, \theta_i^{\,2m-1-i} d\theta_i$$

$$= \frac{r^{2m}}{2m} 2\pi \prod_{i=1}^{2m-2} \int_0^{\pi} \sin sin\, \theta_i^{\,2m-1-i} d\theta_i$$

$$= \frac{r^{2m}}{2m} 2\pi \prod_{i=1}^{2m-1} 2\pi \left(\prod_{j=1}^{2m-1-i} \frac{2j}{2j+1} \right) \left(\prod_{j=1}^{2m-i} \frac{(2j-1)}{2j} \right)$$

$$= \frac{r^{2m}}{m} \pi * \frac{\pi^{m-1}}{(m-1)!}$$

And we get our final version, closed form formula for the hypervolume of a hypersphere with even dimension as

$$Hypervolume = r^{2m} * \frac{\pi^m}{m!}$$

All that we have left is the odd version. Let $n=2m+1$ so that

$$\prod_{i=1}^{2m-1} \int_0^{\pi} \sin sin\, \theta_i^{\,2m-i} d\theta_i = \int_0^{\pi} \sin sin\, \theta^{\,1} d\theta \int_0^{\pi} \sin sin\, \theta^{\,2} d\theta \ \dots \int_0^{\pi} \sin sin\, \theta^{\,2m-1} d\theta$$

$$= (2) * \left(\frac{\pi}{2}\right) * \left(2 * \frac{2}{3}\right) * \left(\frac{\pi}{2} * \frac{3}{4}\right) * \left(2 * \frac{2}{3} * \frac{4}{5}\right) * \left(\frac{\pi}{2} * \frac{3}{4} * \frac{5}{6}\right) \dots \ \pi \prod_{i=1}^{m} \frac{(2i-1)}{2i} * \left(2 \prod_{i=1}^{m} \frac{2i}{2i+1}\right)$$

And like before take these in pairs of two, this time starting at $\frac{\pi}{2}$ instead of 2, since the last one in a pair will be an odd one. We'll remember to multiply by the last two at the end of the calculation. We get in general

$$\frac{2}{3}\pi, \frac{2}{5}\pi, \frac{2}{7}\pi \dots \frac{2}{2m+1}\pi$$

$$\pi \prod_{i=1}^{m} \frac{(2i-1)}{2i} * 2 \prod_{i=1}^{m-1} \frac{2i}{2i+1} = \frac{2\pi}{2m+1}$$

$$\prod_{j=1}^{m-1}\left(2\prod_{i=1}^{m-1}\frac{2i}{2i+1} * \pi\prod_{i=1}^{m}\frac{(2i-1)}{2i}\right) = \prod_{j=1}^{m-1}\frac{2\pi}{2j+1}$$

$$= \frac{\pi^{m-1}2^{m-1}}{\prod_{i=0}^{m-1} 2i+1}$$

And with the first 2 factored in,

$$= \frac{\pi^{m-1}2^{m}}{\prod_{i=0}^{m-1} 2i+1}$$

Let's substitute as we did last time:

$$\int_0^r r^{2m}dr\int_0^{2\pi} d\theta_{2m}\prod_{i=1}^{2m}\int_0^{\pi}\sin sin\,\theta_i^{\,2m-i}d\theta_i$$

$$= \frac{r^{2m+1}}{2m+1}2\pi\prod_{i=1}^{2m}\int_0^{\pi}\sin sin\,\theta_i^{\,2m-i}d\theta_i$$

$$= \frac{r^{2m+1}}{2m+1}2\pi\prod_{i=1}^{2m} 2\pi\left(\prod_{j=1}^{2m-i}\frac{2j}{2j+1}\right)\left(\prod_{j=1}^{2m-i}\frac{(2j-1)}{2j}\right)$$

$$\frac{r^{2m+1}}{2m+1}2\pi * \frac{\pi^{m-1}2^{m-1}}{\prod_{i=0}^{m-1} 2i+1}$$

$$Hypervolume = r^{2m+1} * \frac{\pi^{m} * 2^{m+1}}{\prod_{i=0}^{m} 2i+1}$$

We can recover the hyperareas from these formulas for the hypervolume in a simple manner. If we use shell integration in 2 and 3 dimensions as our model we can extend it to the higher dimensional models. Since a thin shell can be integrated across its radius to give the volume of a hypersphere the converse should hold as well, the radial derivative of the volume of a hypersphere should give the thin shell that is the hyperarea. This is demonstrated best in the lower dimensional cases where we can recover the known perimeter and area of a circle and sphere respectively. We have

$$\{(n = 2m)\ Hypervolume = r^{2m} * \frac{\pi^m}{m!}\ (n = 2m + 1)\ Hypervolume = r^{2m+1} * \frac{\pi^m * 2^{m+1}}{\prod_{i=0}^{m} 2i+1}$$

And for the circle we have an even case of m=1 so

$$r^{2m} * \frac{\pi^m}{m!}$$

$$= \pi r^2$$

And if we derive it radially we should get the circumference

$$\frac{d}{dr} \pi r^2$$

$$= 2\pi r$$

Next is seeing how it works for the sphere. The sphere is a case of m=1 for an odd n

$$r^{2m+1} * \frac{\pi^m * 2^{m+1}}{\prod_{i=0}^{2m+1} 2i+1}$$

$$\frac{=4\pi r^3}{3}$$

Which can be radially derived to get its surface area.

$$\frac{d}{dr} \frac{4\pi r^3}{3}$$

$$= 4\pi r^2$$

And so applying this to both the even and odd cases will result in

$\{\{(n = 2m)\ Hypervolume = r^{2m} * \frac{\pi^m}{m!}\ (n = 2m) Hyperarea = 2 * r^{2m-1} * \frac{\pi^m}{m-1!}\ \{(n = 2m + 1$

Higher Dimensional Fractals

Fractals are figures that are infinitely self-similar. This can also be called by another name, scale invariance. When you zoom in any part of a fractal at any scale it will look like the other parts of the fractal. Approximate fractals are found in myriads of places in the natural world. This is because natural processes are generally recursive, and because fractals solve many types of maximization and minimization problems. For example consider a tree's roots. Their purpose is generally to maximize the surface area for nutrient and water absorption, and the less material they use to do it in the better it is for the tree. So in other words maximize surface area while minimizing volume and staying connected. The pattern that has biologically served that purpose has been the branching type of pattern seen in trees, cardiovascular systems, and many other areas.

One of the most cited fractals in the mathematical literature is the Koch fractal. The Koch fractal is a 2 dimensional figure that is generated using a simple algorithm. One starts with an equilateral triangle and at every step after replaces the middle third of every line segment with a bump that is an equilateral triangle but smaller. This is graphically represented by Figures 9, 10, 11 and in the infinite limit as Figure 12.

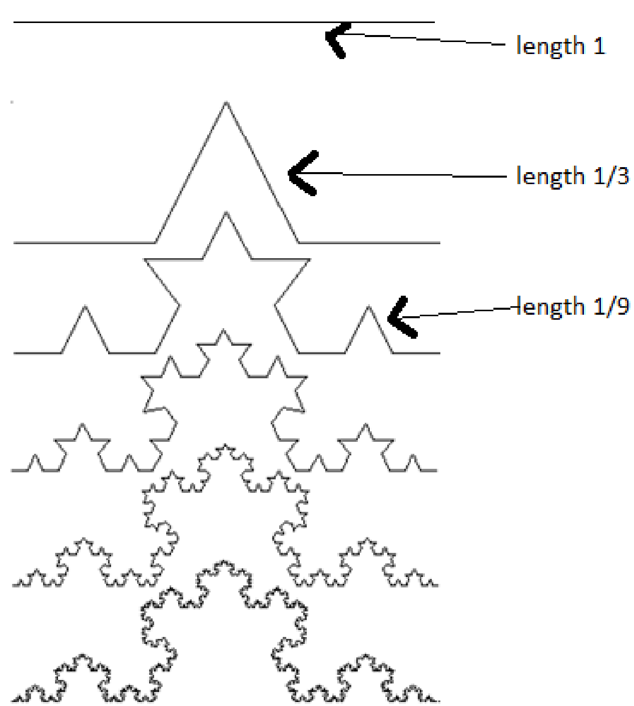


Figure 9: *finite iterations of one side of the Koch curve (Cant et al)*

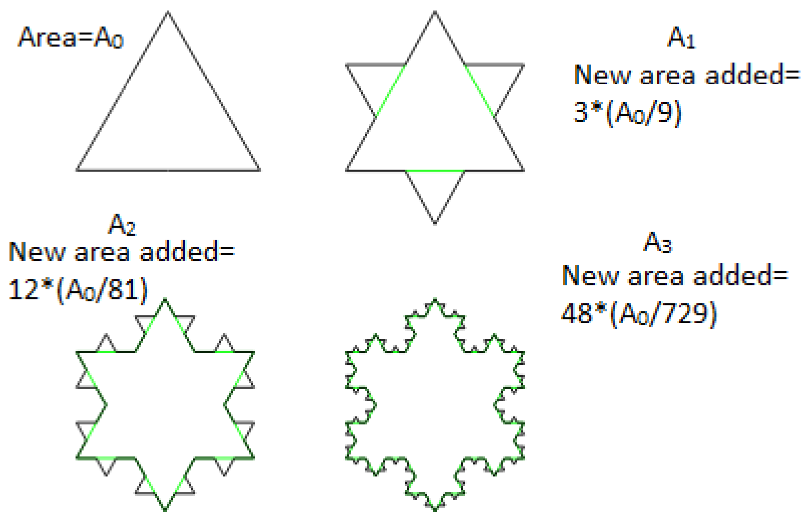


Figure 10: *Considerations of area in finite iterations of the Koch curve (Wxs)*

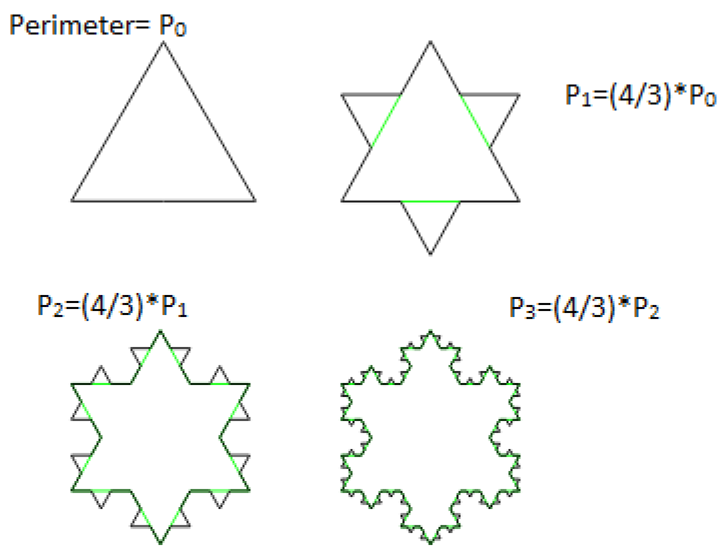


Figure 11: *Considerations of perimeter in finite iterations of the Koch curve (Wxs)*

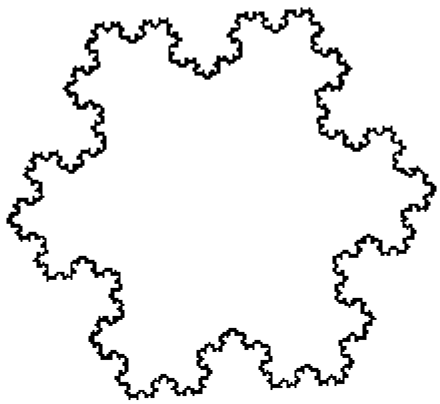


Figure 12: *Infinitely many iterations of the Koch curve (Cant et al)*

Analysis on the Koch curve can be done iteratively. The initial condition has an area of $A_0 = (\sqrt{3}/4) * l$ and a perimeter of $P_0 = 3l$, in which all the other figures can be defined in terms of. It is easier for analysis to divide the perimeter and area to be in terms of the number of segments N_n and the length of each segment L_n. If we look at one side alone we see that the first iteration makes 4 segments of size 1/3 for every one segment of size l. Every iteration does the same set of steps to every equal length and we start at P_0. Thus $N_n = 3 * 4^n$ and then $L_n = (1/3)^n$. Since the perimeter of any figure can be simply represented as $N_n * L_n$ each successive iteration can be represented as:

$$P_n = (4/3)^n * P_0$$

which diverges as n goes towards positive infinity. This is because the numerator is larger than the denominator, and in the infinite limit anything greater than 1 raised to the *n*th power will become infinite without bound.

The calculation for the area is slightly more complicated. One starts with an initial area of A_0 and the first iteration of the fractal results in A_1. To do so A_1 is calculated as A_0 (because the initial area is part of the area of the next iteration) + 3 triangles of area 1/9*A_0. The next step is calculated as A_2=A_1+12 triangles of area 1/81*A_0, as shown by the figure above. It turns out that in general

$$A_n = A_{n-1} + (N_n)/4 * L_n^2 * A_0$$

When we set it up in a geometric series we get

$$A_n = A_0 + \frac{3}{9}A_0 + \frac{12}{81}A_0 + \frac{48}{729}A0... = A_0 * (1 + \frac{3}{4}\sum_{i=1}^{n} (\frac{4}{9})^i)$$

$$= A_0 * (1 + \frac{3}{4} * \frac{4}{9}\sum_{i=0}^{n} (\frac{4}{9})^i)$$

$$= A_0 * (1 + \frac{1}{3}\sum_{i=0}^{n} (\frac{4}{9})^i)$$

which amounts to evaluating a geometric series. In general, for -1<*r*<1

$$\sum_{i=0}^{w} r^i = \frac{(r^{w+1}-1)}{(r-1)}$$

And we can evaluate A_n to be $= A_0 * (1 + \frac{1}{3}\frac{((\frac{4}{9})^{n+1}-1)}{(\frac{4}{9}-1)}$

We can collect the terms with algebra to make it neater

$$= A_0 * (1 + \frac{1}{3}\frac{(1-(\frac{4}{9})^{n+1})}{(1-\frac{4}{9})}$$

$$= A_0 * (1 + \frac{1}{3} * \frac{9}{5}(1 - (\frac{4}{9})^{n+1})$$

$$= A_0 * (1 + \frac{3}{5}(1 - (\frac{4}{9})^{n+1})$$

$$= A_0 * (1 + \frac{3}{5} - \frac{3}{5}(\frac{4}{9})^{n+1})$$

$$= \frac{A_0}{5} * (8 - 3(\frac{4}{9})^{n+1})$$

which converges to (8/5)* A_0 as *n* goes to infinity. This is because in the infinite limit the term $\frac{4^{n+1}}{9^{n+1}}$ has a larger denominator, as n becomes large this will become small.

Another interesting variant of the ordinary Koch curve is the Koch square. In it one starts with a square, and at every iteration replaces the middle third with another square, analogous to the Koch snowflake case. Again, a simpler way of analyzing the system is to divide it into number of line segments and into the length of a line segment. To initialize the system A_0 is equal to the area of the initial square and P_0 is equal to its perimeter. The perimeter at any step along the way can also be represented as $P_n L_n$. For the Koch square if we look at any one side we see that 1 line segment of size l is replaced by 5 line segments of size 1/3. Thus P_n is $4 * 5^n$ and L_n is $(\frac{1}{3})^n$. The perimeter is simply the multiplication of these two and is given by $4 * (5/3)^n$, which diverges as n approaches positive infinity.

Likewise the area is also more complicated for the square case. $A_1 = A_0$+4 squares of area 1/9 A_0. $A_2 = A_1$+20 squares of area 1/81 A_0. The mathematics is closely analogous:

$$A_n = A_0 + \frac{4}{9}A_0 + \frac{20}{81}A_0 + \frac{100}{729}A_0 \ldots = A_0 * (1 + \frac{4}{5}\sum_{i=1}^{n} (\frac{5}{9})^i)$$

$$= A_0 * (1 + \frac{4}{5} * \frac{5}{9}\sum_{i=0}^{n} (\frac{5}{9})^i)$$

$$= A_0 * (1 + \frac{4}{9}\sum_{i=0}^{n} (\frac{5}{9})^i)$$

$$= A_0 * (1 + \frac{4}{9}\frac{((\frac{5}{9})^{n+1} - 1)}{(\frac{5}{9} - 1)}$$

$$= A_0 * (1 + \frac{4}{9}\frac{(1 - (\frac{5}{9})^{n+1})}{(1 - \frac{5}{9})}$$

$$= A_0 * (1 + \frac{4}{9} * \frac{9}{4}(1 - (\frac{5}{9})^{n+1})$$

$$= A_0 * (1 + 1(1 - (\frac{5}{9})^{n+1})$$

$$= A_0 * (1 + 1 - (\frac{5}{9})^{n+1})$$

$$= A_0 * (2 - (\frac{5}{9})^{n+1})$$

Which converges to twice the area of the original square as n goes towards positive infinity. The same arguments for convergence apply as they previously did.

There are other variations possible in addition to looking at other 2 dimensional polygons like the pentagon and the hexagon. It is possible to extend the idea so as to create fractals of this sort in higher dimensional spaces. One example is given by the Koch cube. It is created in an analogous manner, starting with a cube and then proceeding to add a smaller cube in the middle of each square face. This can be represented graphically (one face for simplicity) as in Figure 13:

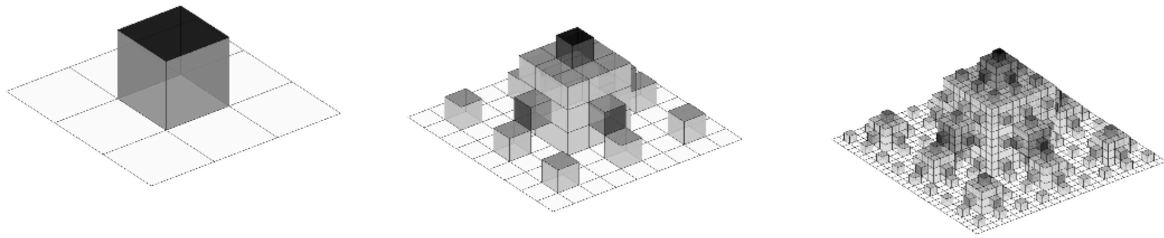


Figure 13: *Finite iterations of Koch Cube (Jacquenot)*

Where before we counted the number of line segments and multiplied by the size of each to get the perimeter and multiplied the number of squares by the area of each square to get the total area, we are looking for two different measures here. We are looking for the area and the volume. If we try to measure how many lines we'd be baffled where to start since even before we got to the first iteration the square sides of the cube have infinitely many lines running between the edges.

Again, as previously, it helps to break down what is being looked at for elucidation of analysis. We need to figure out N_n the number of squares at iteration *n*, and also L_n the length of an edge of the cube. It seems intuitively obvious that the area at iteration n of the Koch cube will be simply $N_n * L_n^2$. When we look graphically at one side of the Koch cube and iterate it one time we see one large square becoming 8 squares surrounding with one cube in the middle. Since one of the cubes faces are on the inside it is not counted and we can get the number of squares created from a previous square as 8+5=13. Since this extends, i.e. the second iteration will have the same analysis for each individual square, $N_n=6*13^n$ and $L_n=1/3^n$. So, the area at iteration *n* is $A_n=6\,(13/9)^n$ which diverges as *n* approaches positive infinity.

Volume is less simple, continuing the pattern. We can compute the volume recursively with the same steps and computations as last time. The initial volume is denoted as V_0; $V_1 = V_0$+6 cubes of size 1/27 V_0; V_2 similarly is V_1+78 cubes of size 729 V_0. The full recursive version is

$$V_n = V_{n-1} + \frac{N_n}{13} * L_n^3$$

$$V_n = V_0 + \frac{6}{27}V_0 + \frac{78}{729}V_0 + \frac{6}{13}\frac{13}{27}^3 V_0 ... = V_0 * (1 + \frac{6}{13}\sum_{i=1}^{n} (\frac{13}{27})^i)$$

$$= V_0 * (1 + \frac{6}{13} * \frac{13}{27}\sum_{i=0}^{n} (\frac{13}{27})^i)$$

$$= V_0 * (1 + \frac{2}{9}\sum_{i=0}^{n} (\frac{13}{27})^i)$$

$$= V_0 * (1 + \frac{2}{9}\frac{((\frac{13}{27})^{n+1} - 1)}{(\frac{13}{27} - 1)}$$

$$= V_0 * (1 + \frac{2}{9}\frac{(1 - (\frac{13}{27})^{n+1})}{(1 - \frac{13}{27})}$$

$$= V_0 * (1 + \frac{2}{9} * \frac{27}{14}(1 - (\frac{13}{27})^{n+1})$$

$$= V_0 * (1 + \frac{3}{7}(1 - (\frac{13}{27})^{n+1})$$

$$= V_0 * (1 + \frac{3}{7} - \frac{3}{7} * (\frac{13}{27})^{n+1})$$

$$= \frac{V_0}{7} * (10 - 3 * (\frac{13}{27})^{n+1})$$

which will converge to 10/7 V_0 as n approaches positive infinity.

But besides 3 dimensional cubes there exist cubes of higher dimension (hypercubes, discussed previously) that are also good for fractal analysis. If we take the same starting postulates we can create higher dimensional fractals of the Koch sort for hypercubes of any dimension. The beginning step is simple, we start with a pristine hypercube with no branches or sub-hypercubes added to it. Then in the first iteration we add a smaller version of the hypercube to each side. In the second iteration we add even smaller copies in more amounts in a way dictated by mathematical method.

To be more precise in this discussion I will divide the analysis into N_n and L_n again for clarity. The first step always starts with looking at one side of a hypercube of dimension *M* to see how many hypercubes of dimension *M-1* it creates after one iteration. With squares we looked to see how many lines it created, with the cube we looked for how many squares it created and this pattern extends on. With the square case we started with one side, one line segment, and it created 5 smaller segments in its stead. This is because the line segment was split into thirds, the middle third being replaced by a square with one side missing (the side touching the interior) so

$$3^1 + 2(2 - 1) = 5.$$

When we analyzed the cube we started with a single square side, turned it into a grid of 9 with the middle missing, which was replaced with a cube with one side missing as well. In this case the numbers are

$$3^2 + 2(3 - 1) = 13$$

This argument extends to the higher dimensional versions as well. When on a cube of dimension M one side of it will be replaced by a grid which divides it into thirds based on every dimension possible to that side, with the middle missing. This is given by

$3^{M-1} - 1$. We must remember too that the middle is replaced with a hypercube of dimension *M* which is smaller and missing one of its sides. The number of *M-1* dimensional figures in a *M* dimensional cube is given by the previous analysis in Chapter One and is 2**M*. When we factor in that one side is missing and add in all the non-center places we get

$$3^{M-1} - 1 + 2 * M - 1$$

or written in a better form

$$3^{M-1} + 2 * (M - 1)$$

This again is the number of *M-1* dimensional facets that one side of an M dimensional figure will split into after one iteration. To get N_n then we must remember to multiply it by the number of faces in an *M* dimensional hypercube. Since the iterations apply to each face on every iteration the full form of N_n is

$$N_n = 2M * (3^{M-1} + 2(M - 1))$$

In each previous example we let the length of an edge of any cube or hypercube on iteration n, L_n, be $(1/3)^n$. This will hold for the following discussion as well.

Computing the hyperarea (HA_n) will be analogous to the discussion of perimeter for the square and for area in the 3 dimensional cube. Simply put we will multiply the number of *M-1* dimensional figures by the area measure each will have. This amounts to

$$HA_n = N_n * L_n^{M-1}$$

$$= 2M * \left(\frac{(3^{M-1}+2*(M-1))}{3^{M-1}}\right)^n$$

which will diverge as n approaches positive infinity because

$$3^{M-1} + 2 * (M - 1) > 3^{M-1}$$

Computing the hypervolume is computationally analogous. HV_0 is the initial hypervolume. In the first iteration $2M$ many figures will be added of volume $1/3^M HV_0$. In the second iteration $2M * (3^{M-1} + 2 * (M - 1))$ many figures of volume $1/3^{2M}$ will be added. In general

$$HV_n = HV_{n-1} + \frac{N_n}{3^{M-1}+2(M-1)} * \frac{1}{3^{nM}}$$

$$= HV_0(1 + \frac{2M}{3^{M-1}+2(M-1)} \sum_{w=1}^{n} (\frac{3^{M-1}+2(M-1)}{3^M})^w)$$

$$= HV_0(1 + \frac{2M}{3^{M-1}+2(M-1)} \frac{3^{M-1}+2(M-1)}{3^M} * \sum_{w=0}^{n} (\frac{3^{M-1}+2(M-1)}{3^M})^w)$$

$$= HV_0(1 + \frac{2M}{3^M} * \sum_{w=0}^{n} (\frac{3^{M-1}+2(M-1)}{3^M})^w)$$

$$= HV_0(1 + \frac{2M}{3^M} * \frac{1-(\frac{3^{M-1}+2(M-1)}{3^M})^{n+1}}{1-\frac{3^{M-1}+2(M-1)}{3^M}}$$

$$= HV_0(1 + \frac{2M}{3^M} * \frac{1-(\frac{3^{M-1}+2(M-1)}{3^M})^{n+1}}{\frac{3^M}{3^M}-\frac{3^{M-1}+2(M-1)}{3^M}}$$

$$= HV_0(1 + \frac{2M}{3^M} * \frac{3^M}{3^M-3^{M-1}-2(M-1)} * (1 - (\frac{3^{M-1}+2(M-1)}{3^M})^{n+1})$$

$$= HV_0(1 + \frac{2M}{3^M-3^{M-1}-2(M-1)} * (1 - (\frac{3^{M-1}+2(M-1)}{3^M})^{n+1})$$

$$= HV_0(\frac{3^M-3^{M-1}-2(M-1)}{3^M-3^{M-1}-2(M-1)} + \frac{2M}{3^M-3^{M-1}-2(M-1)} - \frac{2M}{3^M-3^{M-1}-2(M-1)} * (\frac{3^{M-1}+2(M-1)}{3^M})^{n+1})$$

$$= \frac{HV_0}{3^M-3^{M-1}-2(M-1)} (3^M - 3^{M-1} - 2(M - 1) + 2M - 2M * (\frac{3^{M-1}+2(M-1)}{3^M})^{n+1})$$

$$= \frac{HV_0}{3^{M-1}(3-1)-2(M-1)} (3^{M-1}(3 - 1) + 2 - 2M * (\frac{3^{M-1}+2(M-1)}{3^M})^{n+1})$$

$$= \frac{HV_0}{2*3^{M-1}-2(M-1)}(2 * 3^{M-1} + 2 - 2M * (\frac{3^{M-1}+2(M-1)}{3^M})^{n+1})$$

which converges to, for all M,

$$= \frac{2*3^{M-1}+2}{2*3^{M-1}-2M+2} * HV_0$$

So it continues the pattern in that it will have a bounded hypervolume but will have a divergent hyperarea.

Rubik's Hypercubes

In this chapter I introduce a well-known puzzle, the Rubik's cube. Discussed next are variants changing the numbers of slices in the 3 dimensional case, as well as analogues in higher dimensions. A structure is given for the 3 dimensional cases, and extended to the general case. It is shown what corresponds to moves on the regular Rubik's cube in this notation, and this is generalized to moves on Rubik's hypercubes. Since these cubes have an even parity no two cycle exists swapping only two stickers. Thus the smallest allowable nontrivial change in stickers is a 3 cycle. One can induct 3 cycles of pieces to *D+1* dimensions assuming one has a 3 cycle in *D* dimensions.

For ease of reference a list of the variables denoted in this paper is as follows:

D	the dimension of the cube
s	the number of slices on the cube
S	a hypersurface

d	a geometric d dimensional hypercube
W	the number of stickers on a particular piece

The standard Rubik's cube is composed of 1 cube sliced into 3 layers in height, length, and width, giving 27 cubes. One of these is on the inside, so only 26 in practice are used, composed of 6 center pieces, 12 edges and 8 corners. The centers are covered by one sticker, the edges by two stickers, and the corners by three stickers. The puzzle is considered solved when all the stickers on all sides are the same color. This corresponds to a minimum of 24 solutions, each whole cube rotations of the other. (Take any arbitrary face, there are 6 sides you can place it on, and 4 ways to rotate keeping it on the same side.)

We can extend this puzzle to other 3 dimensional variants by changing the number of slices the cube is divided into in each direction. For example with only 2 slices we have a "baby Rubik's" consisting of only 8 corners. Or, we can go one up and make 4 slices, which gives us 24 centers, 24 edges, and still 8 corners. No matter how many slices, in 3 dimensions a cube will only ever have 8 corners, as intuitively expected. A more rigorous mathematical justification is wanted to explain this invariant.

The cubed numbers describe the number of pieces that would exist if we were to look at the entirety of the cube, inside and out. In practice however with these puzzles we see only the outside surface, and thus will have (for N, an arbitrary integer greater than 1) N^3- $(N\text{-}2)^3$ total pieces. Thus we cut out the inside of the cube with the N-2 term. It can also be seen intuitively with want of better justification that there will be $6(s\text{-}2)^2$ centers, $12(s\text{-}2)^1$ edges, and $8(s\text{-}2)^0$ corners in 3 dimensions.

There exists a formula for computing how many *d* dimensional components are present in the regular *D* dimensional cube. It relies primarily on the analysis given in Chapter One on hypercubes: a *D* dimensional cube can be codified as

$$\sum_{i=0}^{D} \binom{D}{i} y^i 2^{D-i}$$

which is just another way of saying the number of *i* dimensional pieces is $\binom{D}{i} 2^{D-i}$.

However we do not generally refer to *d* dimensional figures on Rubik's cubes, we refer to corners, edges, centers. In other words we refer to them by the number of stickers they have. The relation (which can be taken as axiom or solved by reference to the 3d and 2*d* cases) is that pieces with w stickers in dimension *D* correspond to *D-w* dimensional figures. Assuming 3 dimensions, points (0*d* figures) correspond to corners and are 3 stickered figures; line segments (1*d* figures) correspond to edges and are 2 stickered figures; squares (2*d* figures) correspond to centers and are 1 stickered pieces. In a 4 dimensional space the 0*d* figures correspond to 4 stickered pieces, and likewise the general result comes from similar considerations.

However we have omitted an important part, variants that have differing numbers of divisions per direction. For any line segment sliced into n segments, there will be two on the outside; giving us *n*-2 middle slices, and a constant 2 outside segments no matter what *n* is used. This applies to all line segments in a very general sense. Lines will vary linearly with increased divisions, squares will increase with the square of divisions (since they are divided across twice) cubes vary in a cubed fashion and *n* dimensional hypercubes will vary with a power of *n* to increased divisions. In these simple cubes each side is similar to the other in that they will all have the same number of divisions. Thus the 0*d* points cannot be split up, the 1*d* lines can be split once along their direction, the 2*d* planes can be split in 2 directions, and so on until *D* dimensional surfaces being split along all *D* of their dimensions.

If s is the number of divisions per side, the puzzle has a type of form s^D. This cannot be the final form however for it gives the wrong results, predicting that there will be more pieces than there are on any cube. We can use this idea to generalize the results in Chapter One for a new generating function. Whereas before we had a generating function of

$$(y + 2)^D = \sum_{i=0}^{D} \binom{D}{i} y^i 2^{D-i}$$

we can slightly generalize it to suit our purposes. If we want *s* many slices we have

$$((s - 2)y + 2)^D \sum_{i=0}^{D} \binom{D}{i} y^i 2^{D-i}$$

The full final formula for the number of *d* dimensional figures on a *D* dimensional cube cut into *s* slices then is:

$$2^d \cdot \binom{D}{d} \cdot (s-2)^{(D-d)}$$

(We can substitute *d* dimensional figures interchangeably with *D-d* stickered pieces.) Figure 14 illustrates the number of pieces with *w* stickers in a puzzle of dimension *D*, with 3 slices. Figure 15 illustrates holding *D* at a constant 4 and varying *s*.

		w											total	total
		0	1	2	3	4	5	6	7	8	9	10	pieces	stickers
D	0	1											1	0
	1	1	2										3	2
	2	1	4	4									9	12
	3	1	6	12	8								27	54
	4	1	8	24	32	16							81	216
	5	1	10	40	80	80	32						243	810
	6	1	12	60	160	240	192	64					729	2916
	7	1	14	84	280	560	672	448	128				2187	10206
	8	1	16	112	448	1120	1792	1792	1024	256			6561	34992
	9	1	18	144	672	2016	4032	5376	4608	2304	512		19683	118098
	10	1	20	180	960	3360	8064	13440	15360	11520	5120	1024	59049	393660

Figure 14: *Varying D holding s=3*

Number of pieces with w stickers, for s slices in 4 dimensions

		w					total	total
		0	1	2	3	4	pieces	stickers
	0							
	1							
	2	0	0	0	0	16	16	64
	3	1	8	24	32	16	81	216
	4	16	64	96	64	16	256	512
s	5	81	216	216	96	16	625	1000
	6	256	512	384	128	16	1296	1728
	7	625	1000	600	160	16	2401	2744
	8	1296	1728	864	192	16	4096	4096
	9	2401	2744	1176	224	16	6561	5832
	10	4096	4096	1536	256	16	10000	8000

Figure 15: *D=4 varying s*

Thus, any *n* dimensional object will be bounded by surfaces that are *n*-1 dimensional. We live in a three dimensional world, but all we can ever see are the 2 dimensional surfaces enclosing a structure!

Rubik's cubes have faces covered in stickers. One can use a piece notation to denote the objects under discussion, but in general they lack all the possible degrees of freedom. A piece can be in the same location but rotated or flipped around. The objects being used are collections of vectors; these are assigned to denote one and only one sticker. The way I will notate the entire object is by using set of vectors with multiple indices (one of the indices is special, it denotes the surface). Specifying all the indices will single out a lone sticker on a particular face which has a corresponding unique scalar.

One needs a coordinate system for labeling the stickers. In any *D* dimensional space, cubes occupying it maximally, i.e. using all dimensions, will have 2*D* many *D*-1 dimensional surfaces covering it. A 3 dimensional cube will have 6 two-dimensional surfaces covering it, as shown in the previous analysis in chapter one. Each surface furthermore, in *D* dimensional space, will have $s^{(D-1)}$ stickers contained within it. Each surface of a 3*D* cube with *s* divisions will have s^2 stickers.

Each cube has 2*D* many *D*-1 dimensional surfaces. On each of these surfaces there are s slices per each dimension we have, and so we multiply this out, *D*-1 times, and the total number of stickers is $2D*s^{(D-1)}$

Rubik's hypercubes can be implemented with the aid of a computer program. To that end the following discussion will have to utilize a specific implementation for denoting the structure of a *D* dimensional Rubik's hypercube. The first index denotes which *D*-1 dimensional surface we are looking at on the *D* dimensional cube. All pieces with the same value of *S* will be the same color, since they are on the same surface. The reason I put emphasis on a different label is that all the other parameters will go from 1 to *s*, whereas *S* does not; *S* ranges from 1 to 2*D*, the number of surfaces on a *D* dimensional hypercube.

I want to make a labeling system that retains as much intuition as possible. The way we read a book is from left to right, upon finishing a row we move from top to bottom, and when turning pages, from front to back. If we had a 4 dimensional book, in a 4 dimensional land, then we would turn to the next 4d sheet, and repeat over and over, until we were at the down side of that 4th dimension! So, define *S* as follows: 1 is the left surface, 2 is right, 3 is top , 4 bottom, 5 front, 6 back, 7 up in 4d direction, 8 down in 4d direction, 9 up in 5d direction, and so forth. 2*N*-1 and 2*N* for refer to respectively forward and backward directions along a line in that *N*th dimension. However this is still too crude a system, because on each surface one needs *D*-1 additional numbers to specify a specific *D*-1 dimensional surface sticker. So, let us use *D*-1 indices on each surface and let the value run from 1 to *s*. The particular case of the familiar 3 dimensional Rubik's cube is thus:

$C^{S}{}_{n1,\, n2}$

Specifying each face, this becomes

$C^{1}{}_{bc}$ $C^{2}{}_{bc}$

$C^{3}{}_{ac}$ $C^{4}{}_{ac}$

$C^{5}{}_{ab}$ $C^{6}{}_{ab}$

And specifying each sticker creates six two dimensional arrays:

$C^{1}{}_{11}$ $C^{1}{}_{12}$ $C^{1}{}_{13}$ $C^{2}{}_{11}$ $C^{2}{}_{12}$ $C^{2}{}_{13}$

$C^{1}{}_{21}$ $C^{1}{}_{22}$ $C^{1}{}_{23}$ $C^{2}{}_{21}$ $C^{2}{}_{22}$ $C^{2}{}_{23}$

$C^{1}{}_{31}$ $C^{1}{}_{32}$ $C^{1}{}_{33}$ $C^{2}{}_{31}$ $C^{2}{}_{32}$ $C^{2}{}_{33}$

$C^{3}{}_{11}$ $C^{3}{}_{12}$ $C^{3}{}_{13}$ $C^{4}{}_{11}$ $C^{4}{}_{12}$ $C^{4}{}_{13}$

$C^{3}{}_{21}$ $C^{3}{}_{22}$ $C^{3}{}_{23}$ $C^{4}{}_{21}$ $C^{4}{}_{22}$ $C^{4}{}_{23}$

C^3_{31} C^3_{32} C^3_{33} C^4_{31} C^4_{32} C^4_{33}

C^5_{11} C^5_{12} C^5_{13} C^6_{11} C^6_{12} C^6_{13}

C^5_{21} C^5_{22} C^5_{23} C^6_{21} C^6_{22} C^6_{23}

C^5_{31} C^5_{32} C^5_{33} C^6_{31} C^6_{32} C^6_{33}

Here the indices a, b, and c refer to left-right, top-bottom, and front-back respectively, and can go from 1 to 3. Note especially that faces 1 and 2 are missing index a, that faces 3 and 4 are missing b, and that faces 5 and 6 are missing c. This is because once on the surface of a face one can go no farther in that direction. In other words you can't go any more left if constrained to the left face, you are as left as allowed. For example this means that index c will be the second index for faces 1, 2, 3, 4. Faces 5 and 6 omit it and all the rest will have c as the third index. Fig. 16 illustrates the cube, and its unfolding.

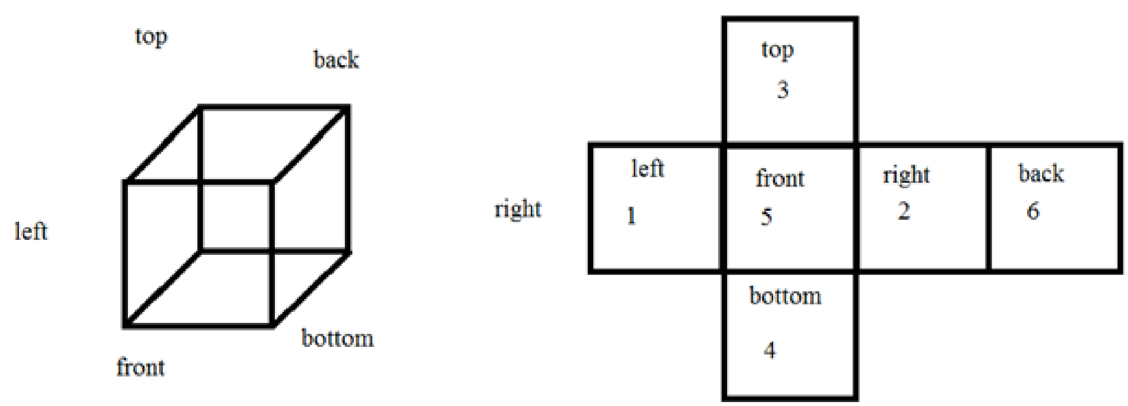


Figure 16: *Unfolding of a Rubik's cube*

For convenience one can assign a scalar based on S and each index p_i (i ranging from 1 to D-1) by the formula

$$(S\text{-}1)*s^{(D\text{-}1)} + p_1+s*p_2+ s^2 *p_3+\ldots\ s^{D\text{-}2} *p_{D\text{-}1}$$

Or more compactly as

$$(S - 1) * s^{(D-1)} + \sum_{i=1}^{D-1} p_i * s^{(i-1)}$$

This will create a set of numbers from 1 to $2D*s^{D\text{-}1}$ and is useful for representing concretely how the labeling should be done. This lets us uniquely label each sticker, even if it changes its position. When one does movements on the cubes, numbers move around. It is very useful for programming purposes to be able to keep separate the particular sticker number and coordinate number. Figure 17 shows how one would label each sticker in the original Rubik's cube with its corresponding scalar, and the order of the indices. This directly corresponds with the previous figure.

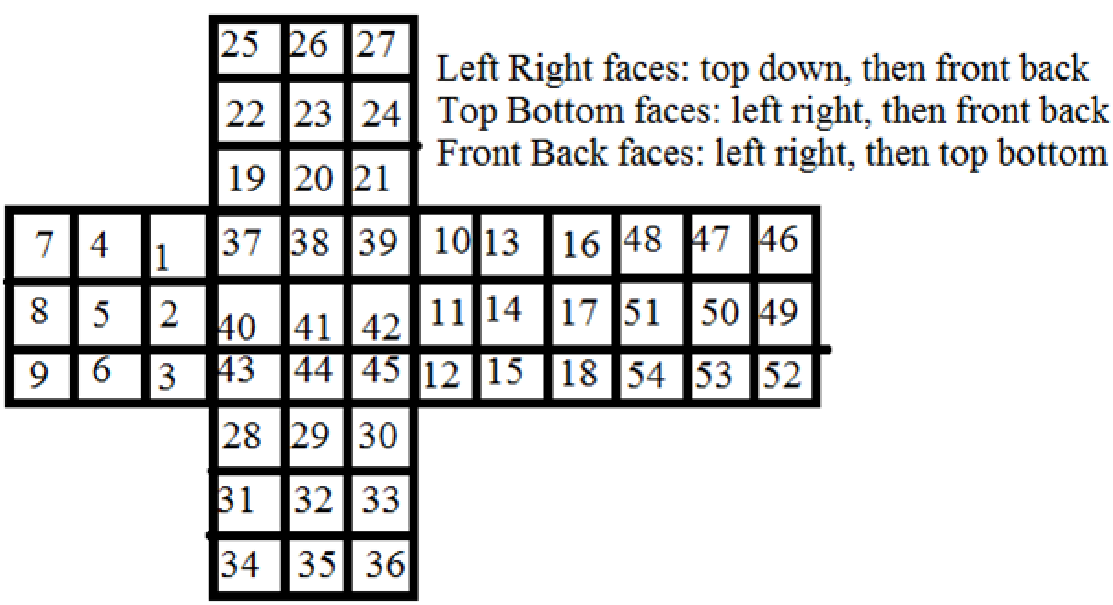


Figure 17: *Numerical Labelling of a Rubik's Cube*

The general and concise way of representing a D dimensional cube would be with $2D$ D-1 dimensional arrays, compacted down into a D dimensional vector. There is only one special index S and we raise it and it ranges from 1 to $2D$, and all the rest of the indices are not special and range from 1 to s:

$$C^{S}_{n1,n2,\ldots nD-1}$$

Note that each sticker lies on exactly one face; a piece consists of stickers that always move together i.e. stickers do not slip off their respective pieces, and are touching; a piece in D dimensions can have from 1 to D stickers.

There is a common notation for the ordinary Rubik's cube that is composed of specifying a face to be turned, and then specifying whether you wish to turn it clockwise or counterclockwise. It turns out that it isn't sufficient for expression on higher dimensional variants. This is because it doesn't have a letter to represent the new faces created by the new dimensions, and because in higher dimensions cubes can be rotated in more ways than simply clockwise or counterclockwise.

I have for this purpose created a new notation for movements on a hypercube. A general movement is a set of 4 integer valued parameters: Face, Axisfrom, Axisto, and Slice.

The range on these parameters is as follows from the Octave program representation: Face in essence refers to which face you are moving if slice is 1. If slice is not 1 then for a given Face there are parallel slices which will be moved instead. Axisfrom tells the starting axis, and Axisto tells where it goes. If Axisfrom and Axisto are switched, the inverse of the move is done. Slice tells how many parallel slices away from the original face one moves. If Slice=1 the move directly touches the face.

Movements on the original Rubik's cube are fairly straightforward, either one is moving an outside face turn or an inner slice turn. One has a choice of 6 different faces and each face can move either clockwise or counterclockwise around only one axis. To denote a move on one of the left face slices (outside or inside) we fill in the first parameter as 1, to tell it to rotate the left face's bottom to its back we will fill in 4 and 6 respectively; next, we specify Slice, in this case we leave it 1. The following notation is the cycle notation, parameterized by the variables b and c.

$M_{1, 4, 6, 1}\,[C^{S}_{n1, n2}] =$

$(C^{1}_{b, c}\,;\,C^{1}_{c, s+1-b})$

$(C^{2}_{b, c}\,;\,C^{2}_{b, c})$

$(C^{3}_{1, c}\,;\,C^{5}_{1, s+1-b}\,;\,C^{4}_{1, s+1-c}\,;\,C^{6}_{1, b})$

To give a more general example, consider $M_{1, 4, 6, slice}$ on a D dimensional cube. The numbers are such that we can have the indices in order without any ellipsis in front or middle.

If slice=1

$(C^{1}_{b, c, d \ldots}; C^{1}_{c, s+1-b, d\ldots})$

If slice >1

$(C^{1}_{b, c, d \ldots}; C^{1}_{b, c, d\ldots})$

Regardless of slice,

$(C^{2}_{b, c, d \ldots}; C^{2}_{b, c, d\ldots})$

$(C^{3}_{slice, c, d \ldots}; C^{5}_{slice, s+1-b, d \ldots}; C^{4}_{slice, s+1-c, d \ldots}; C^{6}_{slice, b, d\ldots})$

All the rest of the extra faces move in this way

$(C^{7}_{slice, b, c, d} \ldots; C^{7}_{slice, c, s+1-b, d\ldots})$

$(C^{8}_{slice, b, c, d \ldots}; C^{8}_{slice, c, s+1-b, d\ldots})$

…

$(C^{2D-1}_{slice, b, c, d \ldots}; C^{2D-1}_{slice, c, s+1-b, d\ldots})$

$(C^{2D}_{slice, b, c, d \ldots}; C^{2D}_{slice, c, s+1-b, d\ldots})$

For derivation of why the extra faces would do this, think about the fact that no sticker can slip off of its piece. For any face, pick a corner sticker. Find out what other stickers are attached to that corner, and in order to satisfy the no sticker slippage constraint we note that the sticker will stay on the same face, but will have to move to a position that is touching where the corner has moved to on its own face.

To be quite general, if we do one of the same moves like rotating the top face clockwise, but on a cube of higher dimension, many of the same things will happen. The entire top face rotates around (although the nature of the face is of a higher dimension). A part of the left side goes to the back, which goes to the right, which goes to the front. The only problem then is discovering what will happen to the new faces that appear on the higher dimension analogues. We can deduce rather easily that only a part of it will move, ie that when we move the top face the new faces do not move in entirety like the top, but only a portion of it, and that portion will be the slice closest to the top.

As an example consider the 4 sliced 5 dimensional case of sticker $1_{1234.}$ (The unique scalar associated with this sticker is $(1-1)*4^4+ 1+2*4+3*4^2+4*4^3$ or 0+1+8+48+256=313.) To establish touching we will note that in our 3 dimensional cube for each index that is an extremum, either 1 or s on a cube with s slices there is a sticker in that direction that is touching. Likewise here we first note that since we are on side 1 we cannot move in that direction so our first index corresponds to movement along the 3-4 line. In this case it is 1 and we have a sticker on the 3 face somewhere touching ours. On the next 5-6 line we have a 2 and so it is not touching either of those faces but lies in between both. Next is 7-8 and we have a value of 3, also in between. Lastly is the line 9-10 and we have a value of 4. Since our s here is 4 we know it is touching the even one out of the pair of faces, 10. From this we see that 1_{1234} is on face 1, and is touching faces 3 and 10, and that this will be a 3 stickered piece on a 5 dimensional hypercube.

On face 3 how do we find a sticker that is touching 1_{1234}? Since it has to be touching at least face 1 we know its first index (corresponding to 1-2 line) must be 1 ie $3_{1???}$. On 1_{1234} the 2[nd] index (referring to 5-6) is 2, in order for our sticker on face 3 to match up in the direction it too must be 2. Likewise for the other directions that they share in common, they all must match up in order to be touching. Thus, we have 3_{1234}. It is left as an exercise to the reader to verify that 10_{1123} is the third sticker.

The most general form for movements on a D dimensional cube is

$M_{\text{Face; Axisfrom; Axisto; Slice}}$

It has been shown that the Rubik's cube has even parity. This means that any set of moves will have an even number of 2 cycles. A single 2 cycle swaps two pieces (Unknown Author MIT.edu). Since the cube is even there exists no single 2 cycle like (1, 2). Thus nondegenerate sets of 2 cycles will move at least 3 pieces, such as (1, 2)(2, 3) = (1, 2, 3) Therefore, the smallest possible move sets (not counting the null movement) move at a minimum 3 pieces.

The main essence of this argument line is that one can systematically start at a "first" 1 stickered piece and put the right piece there, work on the next one being careful not to disturb the one already solved, and continue until all 1 stickered pieces are solved. Then applying the same principle to 2 stickered pieces and all the rest, Rubik's puzzle of any dimension can be solved systematically by these smallest sets of moves.

For simplicity of notation assume that all the pieces lie on one of the slices of the 1 (left) plane. That this is general enough to encompass all cases considering that moving the entire cube until the desired face is on the left will give the same result. To have the same stickered pieces means that the same number (w) of faces have to be specified, and to qualify as a piece they have to be touching. In order for that to be satisfied whenever an odd face is specified with the current face, the parameter corresponding to it has to be 1. (Because it is the furthest in that direction to touch the face). Likewise, any even faces specified in conjunction with the face at hand has the corresponding parameters set to *s*. For a w stickered piece in *D* dimensions there are faces, each with *D*-1 many free indices which have to mesh up with each other by transformation of $M_{1, 4, 6, n}$. This means that the indices notating for directions 2 and 3 have to match up with the one in front of it after switching and (*s*+1-*c*)-ing it. All the others are not changed and so are constrained to be exactly the same. Another, and possibly the most important constraint is that when one actually does the move, the 3 that are in the cycle undergo a 4 cycle with one more piece.

Here I use a much more simplistic notation for clarity. We can talk about 4 pieces that have *w* stickers apiece, as pieces 1, 2, 3, and 4 (having *w* faces specified which in turn all have their parameters listed and in accordance with said constraints) We assume to possess a set of moves (which will be referred to in shorthand as Move) in our *D* dimensions that moves 3 of these *w* stickered pieces like so:

(3,1,2,4)

But on that face or possibly the slice there exists a move: $M_{1, 4, 6, n}$ that will permute all 4 pieces, and a few others that are around it

(312)(4321) and (other disjoint pieces moved)

The few others are something to worry about but we take care of it later by ensuring they are moved back. Our move in *D*+1 will be the exact same as before, except there is now an additional set of indices, and 2 new faces. The additional set of indices means that the 3 cycle (123) is now s many 3 cycles

$(1_e 2_e 3_e)$ in other words from $(1_1 2_1 3_1)$… to $(1_s 2_s 3_s)$

The additional set of indices also means that our move $M_{1, 4, 6, n}$ moves all *s* many of the 3 cycles in a 4 cycle with other pieces, analogous to before. Luckily though, we do not use $M_{1, 4, 6, n}$ rather we use $M_{2D+1, 4, 6, m}$. Why? Let us examine. When we use one of the 2 new faces around the same axes we ensure that face 1 is neither any of the 4 picked by the Axisfrom and Axisto (see move definition), nor is it the face whose pieces all move, so face 1 is one of the extras who move only a thin layer, the parameter corresponding to the *D*+1th direction being specified by slice being = m:

$$(1_{c,s+1\text{-}b,d\ ,slice=m\ \ldots})$$

When we use $M_{2D+1, 4, 6, m}$ the only stickers that intersect with pieces affected by this move are either attached to face 1 (first index is 1), or are all on a slice across from it (first index=n), parallel. In either case the move will make a 4 cycle out of only a single section (corresponding to the m slice). Still though we have those pesky other pieces that keep on moving. The only way to make all of them go back as if not moved is if we simply undo by doing our move, ie $M_{2D+1, 4, 6, m}$! That is actually what we do, except first undoing our *s* many 3 cycles.

In shorthand, first enact Move, then $M_{2D+1, 4, 6, m}$ then Move^{-1} and $M_{2D+1, 6, 4, m}$. For those versed in group theory, to induct a 3 cycle into a higher dimensional 3 cycle we simply create the commutator of our 3 cycle and a single move. If we expand out Move and $M_{2D+1, 4, 6, m}$ and their corresponding inverses in cycle notation and in the proper order for clarity, we get:

$(3_s 1_s 2_s)$ as Move

$(1_m 2_m 3_m 4_m)$ for a particular m (and pieces invariant of Move) as $M_{2D+1, 4, 6, m}$

$(3_s 2_s 1_s)$ as Move^{-1}

and $(1_m 4_m 3_m\ 2_m)$ as $M_{2D+1,\ 6,\ 4,\ m}$

We can see by composing these cycles that the only pieces affected in going up D+1 are $1_{m,}\ 2_{m,}$ and 3_m which is an ordinary 3 cycle like we hoped for. We set up s many 3 cycles, use a 4 cycle to mess with only a slice, undo those 3 cycles, and then undo the 4 cycle! It puts those messed up ones back, and the only pieces affected overall are the ones that lie at the intersection of the two types of moves.

Magic Hypercubes

This chapter concerns itself with the problem of magic hypercubes. Magic squares are a type of recreational math puzzle where the aim is to create an n x n square where every row column and main diagonal sum to the same number. An example of a 3 x 3 magic square is:

4 9 2 3 5 7 8 1 6

In the most abstract formulation of magic squares we replace every position with a letter and derive relations among them that necessarily hold. This creates a set of linear equations with unknowns. Represented in square form,

$$a\, b\, c\, d\, m\, f\, g\, h\, i$$

And it must hold that

$a + b + c = d + m + f = g + h + i = a + d + g = b + m + h = c + f + i = a + m +$

Which is 8 equations in 10 variables (linearly dependent). Let us add up all the equations that have m to derive an interesting result.

$$d + m + f + b + m + h + a + m + i + c + m + g = The\,Sum * 4$$

We can rearrange the terms to get that

$$a + b + c + d + m + f + g + h + i + 3 * m = The\,Sum * 4$$

$$The\,Sum * 3 + 3 * m = The\,Sum * 4$$

$$3 * m = The\,Sum$$

This result holds not only for the 2 dimensional case but also for the 3 dimensional case, since we can take a middle slice of it and do the same analysis and get the same result, and likewise for any hypercube of any dimension this result holds that the sum is 3 times the middle number.

$1\,1\,1 \quad 1\,1\,1 \quad 1\,1\,1\,1 \quad 1 \quad 1\,1 \quad 1 \quad 1\,1 \quad 1 \quad 1\,1 \quad 1 \quad 1 \quad 1 \quad 1\,1 \quad a\,b\,c\;\, d\,m\,f\;\, g\,h\,i \quad = T$

This matrix represents the equations we mentioned above in a neater form. We can add the 4th 5th and 6th equations together, then subtract the 1st and the 2nd equations to arrive at the third equation, and so throw it out (as a combination of the others) to reduce it to a set of 7 independent equations in 10 variables.

It is easy to prove from that if we define the numbers a m and c then we can derive all the other ones. The form of it will look like this

$$a\, m - a - c\, c\, m + c - a\, m\, a - c - m\, 2m - c\, c + a - m\, 2m - a$$

Rather than a strict linear algebra analysis of the matrices that represent magic hypercubes or magic squares, we deal with them here in a slightly different way, using induction and position vectors to derive a result for the sufficiency of the constants that have to be initialized.

Define a magic 3^n hypercube by a set of n-vectors ($w_1, w_2, \ldots w_n$) where the w_i are integers ranging from 1 to 3, where each vector has a corresponding scalar associated to it (the vector tells the position, the scalar tells the number "in" the box). Elements of dimension 0 will have 0 "2"'s in the vector position, this is because to be a point the vector must contain only 1s and 3s, i.e. a point is on the very edge of every dimension. Likewise elements of dimension f will have f many 2's in their positions. Suppose vector $y_a = (1, y_1, y_2, \ldots y_{n-1})$ has z many 2's and that vector $y_b = (3, y_1, y_2, \ldots y_{n-1})$ also has z many 2's. Then, in between y_a and y_b is $(2, y_1, y_2, \ldots y_{n-1})$, which will have z+1 many 2's. That is to say in between any two elements of dimension z which differ by only one coordinate, is an element of dimension z+1.

A sufficiency theorem about the number of initialized constants can be proven here inductively. A sufficient number of constants to uniquely determine the magic 3^n hypercube is $1+ 2^{(n-1)}$

Proof:

Base case: All elements of dimension 0 (points on the hypercube) are initialized via the following procedure. All points from (1,1…1) to (1,3…3) are given a number $(m+x_i)$ where i ranges from 1 to 2^{n-1}. Also, the centermost point (2,2…2) is given the number m. Since we want the diagonals to sum to the same sum as all the others, this is enough to determine the conjugate points [for (1,1…1) it is (3,3…3), for all the points it is found by replacing all the 3's with 1's and vice versa] which are now $(m-x_i)$(because $(m+x_i)+m+(m-x_i)=3m$).

Inductive step: The values of 2 elements of dimension N which differ by one direction and the sum which every row and column must add to are sufficient to determine the value of the element of dimension N+1 which lies in between them.

We have the entire set of dimension 0 elements from the base case and the sum from the diagonals sum. This is enough to determine the entire set of dimension 1 elements, thus dimension 2 elements, and all elements can be uniquely determined from the selection of $1+ 2^{(n-1)}$ constants.

Conclusion

Now we deal with all the new possible lines of mathematical research this thesis could be extended on. It is possible to extend Chapter One by solving for hyperareas and hypervolumes of mixed figures. It has proven an intractable problem for some time for the author since in general the height and apothem of these figures depend on which operations were performed and in which order. Also possible is an extension of Chapter Two by allowing mixes, cones and prisms of higher dimensions using hyperspheres as the bases. There is probably a method one could use to introduce it as a third operator and insert it into the type of discussion on Chapter One. For example if one makes a four dimensional prism using a cylinder as the base, one will get 4 cylinders, 4 circles and a strange four dimensional figure. Also an extension of Chapter Three is possible by allowing fractals to be done on Koch simplices as well. But like the other possibility in Chapter One we can also do it on mixed figures like square pyramid or triangular prism fractals. All the above fractals were of the Koch sort and so at each iteration gained material. There do exist fractals of reductive form like the Cantor set that are formed by taking away material rather than adding, so one could apply this method to the mentioned hypercubes, simplices and mixed figures. Also, one can do both fractal operations at the same time, adding and removing, or taking alternations, or any of the infinitely many other combinations. Also the size scale was somewhat arbitrarily set at 1/3. It would be a trivial matter of substitution to prove what the hyperareas and hypervolumes will be with a 1/n size scale. It would not however be trivial to ascertain what the case would be if the size scale depended on the iteration, as one no longer obtains a geometric series. I am certain that the analysis on Rubik's hypercubes could be applied to simplices, with some barycentric coordinate system. Many of the same results such as parity theorems would apply in that case as well. Likewise, 3 cycles could be inducted using commutators of lower dimensional three cycles and orthogonal moves. Using mixed figures for Rubik's type analysis is also a possibility. And it is interesting to note that there is more than one solution to these types of puzzles. Since they are color based, for the centermost pieces, the pieces with only one sticker, like the centers of a 3d cube, unlike edges and corners, comprising 2 and 3 stickers respectively, as long as they are on the right surface, it doesn't matter how the centers are oriented with respect to each other, they are indistinguishable. A red center is a red center. For the sake of being able to know which pieces precisely will be moved, I had labeled all as such, but in solutions they can be simplified with this tiddly bit of knowledge. Also, not accounted for is orientation of the centers: some puzzles have a picture on each sticker for each face and requires center orientation, but we did not consider that case. I did not have time to treat the problem of induction of orientation type moves, those that leave all the pieces in the same relative places, but move the stickers around in the same position, like rotating a corner in place. Lastly, a discussion of topology would have been warranted as well. Any of the Koch fractals mentioned would form a basis in their dimension. Hypercubes could be considered with many iterations of the product topology, Simplices with a quotient topology of the hypercube of its dimension, and correspondingly mixed figures as products and quotients at the same time. All the figures of a given dimension have the same genus that is there are no holes, so it seems intuitive to say they are all homeomorphic to one another, triangles to squares to circles, as tetrahedrons are to cubes, triangular prisms, square pyramids, cylinders, cones and spheres, and so on.

This paper has relied on the use of combinatorics, programming, multivariable calculus, advanced algebraic techniques, and analysis of fractals to convey the power and the beauty of the tool that is the concept of multiple dimensions. All these examples have at base relied on the single idea of recursion, of finding a complicated thing in terms of its ancestors, as a key piece. Recursion doesn't only apply to generating higher dimensional equivalents of a mathematical abstraction though. In a very broad sense it is a framework for understanding systems in the most abstract sense of the terms. All non-static systems change over time, and the states of the system are related to the states it had a very short time ago. We can write this recursively using t to denote an arbitrary time and ε as a small time displacement as

$$State(t + \varepsilon) = StateTransformationFunction[State(t)]$$

This is somewhat of the guiding idea behind the *Theories of Everything* championed by physicists. If you can map down what the convoluted function is that relates everything to everything else, and know what everything in the Universe started as, then in principle you could predict anything. Of course, given quantum limitations on knowledge about simultaneous pairs of conjugate variables of a single particle, all one really can get to is a very good probabilistic wavefunction, but often times that is enough to have enormous real world impact. Higher dimensional mathematics are a certain way to better understand many of the abstract underpinnings of science, technology and culture. It is a viewpoint forged under many layers of deep and abstruse mathematical syntax, but the clarity of thought it affords opens many doors.

References

Eaton, J. W. (2013). GNU Octave Software 3.8.2. Free Software Foundation. Retrieved March 1, 2015, from https://www.gnu.org/software/octave/

Jackson, F., & Weisstein, E. W. (n.d.). Regular Tetrahedron. *MathWorld—A Wolfram Web Resource.* Retrieved March 1, 2015, from http://mathworld.wolfram.com/RegularTetrahedron.html

Jacquenot, G. (2010, October 3). Koch Cube Animation Gray. *Wikimedia Commons.* Retrieved March 1, 2015 from http://commons.wikimedia.org/wiki/File:KochCube_Animation_Gray.gif#

Maths for Europe. (March 1999- February 2000). *Fractals.* Retrieved March 1, 2015 From http://mathsforeurope.digibel.be/Leenfrac.html

User wxs. (2007, April 6). KochFlake. *Wikimedia Commons.* Retrieved March 1, 2015 From http://commons.wikimedia.org/wiki/File:KochFlake.svg

Web.MIT. (2009, March 17). *The Mathematics of the Rubik's Cube: Introduction to Group Theory and Permutation Puzzles.* Retrieved March 1, 2015, from http://web.mit.edu/sp.268/www/rubik.pdf

Appendix A

This is the code written in Octave that generates, manipulates, and displays the computational equivalents of higher dimensional Rubik's Cubes. There are 7 files:

1. Testing
2. CubeMove
3. AxisMove
4. FaceMove
5. OtherMove
6. Flip
7. Position

All are runnable in Octave or MatLab so long as the title is removed. Alternatively I can provide the source code if you email me at daniel.kette@gmail.com

Testing is the main program that you will run, all the others are called by this program or by programs ran by it. CubeMove is the main manipulative on the hypercube objects, itself calling AxisMove, FaceMove, and OtherMove. Flip and Position are called by AxisMove, FaceMove and OtherMove and work on the vector representation of the subsets that move.

Testing

```
D=0;
s=0;
while (D<3)
  D=input("enter the number of dimensions \n")
endwhile
while (s<2)
s=input("enter the number of slices or divisions per side \n")
```

```
endwhile
iter=1;
%Creates the Cube Object
Cube={};
                A=[1];
                for i = 1:D-1
                    B=A;
                    for j = 1:s-1
                        A=cat(i,A,B+iter);
                        iter=iter+s^(i-1);
                    end
                end
for k = 1:2*(D)
Cube=[Cube,A+(k-1)*iter];
end
Cube;
in2=[];
%interface with options to
%see the cube                S
%Move the cube               M
%Or to Quit                      Q
in1=" ";
disp ("Hit Q to quit, S to see state of Cube, or M to move it \n")
while (in1!="Q")
 if (in1=="S")
                Cube
 endif
 if (in1=="M")
```

```
for i = 1:4
in2(i)=input("face, then axisfrom axisto and
slice \n")

end

Cube=CubeMove(in2(1),in2(2),in2(3),in2(4),Cube,D,s);
 endif
in1=input("Input", "s");
endwhile
```

CubeMove

```
function [Cube] = CubeMove(Face,AxisFrom,AxisTo,Slice,Cube,D,s)
%portion for the Primary Move of face, given slice=1
%%%%%%%%%%%%%%%%%%%%%%
if (Slice==1)

Cube=FaceMove(Face,AxisFrom,AxisTo,Slice,Cube);
endif
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%
```

```
%portion for the non axis from or to, nor their opposites, nor face, nor face opposite
Cube=OtherMove(Face,AxisFrom,AxisTo,Slice,Cube,D,s);

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%

%portion for axis from and axis to
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%5
Cube = AxisMove(Face,AxisFrom,AxisTo,Slice,Cube,D,s);

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%

end
```

AxisMove

```
function [Cube] = AxisMove(Face,AxisFrom,AxisTo,Slice,Cube,D,s)

iter=1;

% first pull out portions corresponding to the subset
%next permute which face each portion goes to, then
%change the permutation so that the axisfrom has indexed reverse order
```

```
FaceI=(Face+mod(Face,2))/2;
iter=1;
B=[1];
for i = 1:D-2
                                T=B;
                                for j = 1:s-1
                                    B=cat(i,B,T+iter);
                                    iter=iter+s^(i-1);
                                end
end

z=Position(FaceI,s,D,Slice);



C=Cube{AxisFrom};
D1=Cube{AxisTo};
E=Cube{AxisFrom-1};
F=Cube{AxisTo-1};

G=C;
H=D1;
I=E;
J=F;

for i=1:s^(D-2)
                                G(z(i))=F(z(i));
                                H(z(i))=C(z(i));
```

```
                I(z(i))=D1(z(i));
                J(z(i))=E(z(i));
end
C=G;
D1=H;
E=I;
F=J;


fr=AxisFrom/2;
to=AxisTo/2;
                if (AxisFrom>Face)
                    fr=fr-1;
                endif
                if (AxisTo>Face)
                    to=to-1;
                endif


z2=Flip(FaceI,s,D-1,Slice,to-1);
%Flip the moved portion of G and I along their to axis

for i=1:s^(D-2)
                G(z(z2(i)))=C(z(i));
                I(z(z2(i)))=E(z(i));

end
[Cube{AxisFrom}]=G;
```

```
[Cube{AxisTo}]=H;
[Cube{AxisFrom-1}]=I;
[Cube{AxisTo-1}]=J;
```

FaceMove

```
function [Cube] = FaceMove(Face,AxisFrom,AxisTo,Slice,Cube,D,s)


fr=AxisFrom/2;
to=AxisTo/2;
if (Face<AxisFrom)% To ensure the correct parameters are picked out
                                    fr=fr-1;
endif
if (Face<AxisTo)
```

```
                                    to=to-1;
endif
A=Cube{Face};
A=rotdim (A, -1, [fr, to]);
[Cube{Face}] = A;
End
```

OtherMove

```
function [Cube] = OtherMove(Face,AxisFrom,AxisTo,Slice,Cube,D,s)
%portion for the non axis from or to or their opposites, nor face, nor face opposites

iter=1;
B=[1];
                                    for i = 1:D-2
                                        T=B;
                                        for j = 1:s-1
```

```
                        B=cat(i,B,T+iter);
                        iter=iter+s^(i-1);
                end
        end
%That creates another Cubelike object that the rotations are done to
for a = 1:2*(D)

%This for loop goes through every face, deciding whether anything needs
%to be done to it, and if so, what
        fr=AxisFrom/2;%Turning input into correct parameter type
        to=AxisTo/2;
        FaceI=(Face+mod(Face,2))/2;
        if (a<AxisFrom)
                fr=fr-1;
        % The order matters,
        %If we are using Face=2 then the parameter needs to be 1 on
        %a=1 whihc corresponds to face number 1, and for all the others needs
        %to be 2
        endif
        if (a<AxisTo)
                to=to-1;
        endif
        if (a<FaceI)
                FaceI=FaceI-1;
        endif
```

```
if (a!=AxisFrom && a!=AxisTo && a!=AxisFrom-1 && a!=AxisTo-1 && a!=Face)
if((mod(Face,2)==1&&a!=Face+1)||(mod(Face,2)==0&&a!=Face-1))
                        O=Cube{a}
                        disp ("Original Above
\n")

z=Position(FaceI,s,D,Slice)
                        B
                        disp ("B above \n")
                        for i=1:s^(D-2)
                            B(i)=O((z(i)));
                        end
                        B
                        disp ("B after loop
\n")
                        fr=fr-1;
                        to=to-1;
                        z;
                        B=rotdim (B, -1, [fr,
to])%
                        for i=1:s^(D-2)
                            O(z(i))=B(i)
                        end
                        [Cube{a}] = O;

                    endif
                endif
end
```

Flip

```
function [z2] = Flip(FaceI,s,D,Slice,FlipAxis)


iter=1;
B=[1];
for i = 1:D-1
                T=B;
                for j = 1:s-1
```

```
                        B=cat(i,B,T+iter);
                        iter=iter+s^(i-1);
                end
end
y=[];
%makes a vector of length D of all zeroes
for x = 1:D
                y=[y,0];
end
B;
s^(FlipAxis-1);
A=B;
index=1;
i=0;
z2=[];
%this loop creates a vector of the positions to be flipped.
while (index<=D-1)
                i=i+1;
                y;
                ysum=0;
                for n=1:D-1
                        if (n!=FlipAxis)

ysum=ysum+y(n)*s^(n-1);
                        endif
                        if (n==FlipAxis)
                                ysum=ysum+
((s-1)-y(n))*s^(n-1);
                        endif
```

```
                    end
                    ysum;
                    z2(i)=ysum;
                        while (y(index)==s-1)
                                y(index)=0;

index=index+1;

                        end
                    if (index!=D)
                        y(index)=y(index)+1;
                            index=1;
                    endif
end
z2=z2+1;
z2
```

Position

```
function [z] = Position(FaceI,s,D,Slice)


y=[];
%makes a vector of length D of all zeroes
for x = 1:D
                    y=[y,0];
end
y(FaceI)=Slice;
```

```
if (FaceI!=1)
            index=1;
else
            index=2;
endif
i=0;
z=[];
%this loop creates a vector of the positions to be picked out.
while (index<=D-1)
            i=i+1;
            y;
            ysum=0;
            for n=1:D-1
                        ysum=ysum+y(n)*s^(n-1);
            end
            ysum;
            z(i)=ysum;
                        while (y(index)==s-1)
                                    y(index)=0;
                                    if (index+1!=FaceI)

index=index+1;
                                    endif
                                    if (index+1==FaceI)

index=index+2;
                                    endif
                        end
            if (index!=D)
```

```
                    y(index)=y(index)+1;
                    if (FaceI!=1)
                        index=1;
                    else
                        index=2;
                    endif
                endif
end

end
```